

RESEARCH

IOT-Enabled Real-Time Monitoring and Predictive Maintenance of PV-FED BLDC Irrigation Water Pumping Systems

Sevugan Rajesh. J · Revathi. R · Karthikeyan. R · Rajnarayanan. P. N

Received: 24 December 2025 / Revised: 12 April 2026 / Accepted: 28 April 2026

© The Author(s) 2026

Abstract

Photovoltaic (PV) powered BLDC Motors are finding greater applications in present day water pumping systems, especially in the farming areas. Though the use of Renewable energy is an appreciable feature, this system suffers from unintended outages and reduced operational efficiency. This is due to the lack of smart methods to identify and fix the problems before they arise. In addition, the repairing work done post the occurrence of faults is always expensive and there are chances of prolonged degradation. Because of the absence of a real-time monitoring system, there is no provision to alert the user about serious issues. Thus, the aim of this work is to configure a real time setup which can (1) identify the faults using Machine Learning (ML) tools (2) trigger warning messages via Fuzzy Logic control (3) provide greater working strength and live monitoring using IoT. The IoT based problem checking and fixing is made possible through sensors placed on the different parts of the system and the link with the Cloud Storage. Data validation algorithms and cloud-based tools are employed to ensure the accuracy and reliability of sensor data. Based on the learned fault patterns, the smart model classifies potential faults over time. The timely warning messages generated through the IoT system enable effective monitoring and support informed decision-making. The proposed system is developed and evaluated using MATLAB and ThingSpeak platforms through simulation testing. The results indicate that the MPPT accuracy reaches 96.3% even under varying solar conditions. The output efficiency during the test runs is found to be 89.4%. The drop seen in the Fault Rate is 0.75. The pictures from cloud have helped for better visualization and application. For future expansion, edge thinking setups can be considered. Thus, this proposed system guarantees improved irrigation and power usage.

Keywords IoT Monitoring · Predictive Maintenance · BLDC Motor · Photovoltaic System · Water Pumping

1 Introduction

The necessity for sustainable water management has led to the unveiling of water pumping units powered by Photovoltaic (PV) fed Brushless DC motor (BLDC). Though this type of pumping system is greatly helpful for farming locations at remote places, they do suffer from common anomalies. Some of them are inefficient human inspection, unanticipated equipment breakdowns and exorbitant service charges. These problems can be



overcome by including the feature of Internet of Things (IoT), which is an effective technique to continuously observe the system parameters like solar light intensity and water flow rate. Further, it also aids in finding faults at an early stage itself. This IoT feature transforms the system into a smart working model with better efficiency and minimized operating expenses. In order to shed more light on this topic, a detailed literature survey has been carried out pinpointing their merits, research gaps and challenges. To begin with [1], demonstrates the use of Solar powered BLDC motor pump unit, whose DC-DC converter is controlled by an improved Maximum Power Point Tracking (MPPT) technique. Though this combination helps in achieving improved performance and strength, it has failed to accommodate the live observation feature of IoT and early commissioning stages. This issue has been addressed by [2], where IoT and Smart Sensors are employed for PV Powered unit, providing controlled water supply to villages. In spite of IoT inclusion, this method couldn't be able to achieve desired performance, owing to absence of advanced failure detecting techniques, which are essential for system's long life. Work reported by [3] includes multiple sources fed IoT based supply units. In this, there is no clear explanation about the importance solar PV fed BLDC motor pumps and their associated smart features. In contrast to this [4], has proposed neural network based automated water powering system for efficient energy usage. But the possibility of fault prediction in PV powered BLDC motor pumps were not explored.

[5] discusses the effect of employing solar based Permanent Magnet Synchronous Motor (PMSM) drives in electric cars to facilitate energy savings. Though the results were satisfactory, the idea couldn't be extended to water pumping applications. The IoT based fine farming management as reported by [6] effectively collects and processes the data for efficient operation. Still, they could not perform fault identification and rectification in case of PV fed BLDC motor pumps. Similarly, the work demonstrated by [7] for smart water usage using solar power linked BLDC Motor Drive also lacks smart techniques like IoT for early fault diagnosis. Though smart energy saving techniques like Namib Beetle Optimization and Spiking Deep Residual Networks (NBO-SDRN) were proposed by [8] for smart energy control, the absence of IoT tools makes it less capable of identifying the faults. The power sharing based PV fed BLDC motor modelled by [9] doesn't have better repair management owing to the non-inclusion of IoT [10]. proposes a fuzzy based control of PV fed DC Motors, where there is no scope for health monitoring of the system. Smart Energy usage with the help of switched reluctance motors were recommended by [11], where real time monitoring and caring were not considered. The review work presented by [12] highlights the various controlling techniques for the operation of PV based motor pumps, but less importance is given to adaptive monitoring. The role of IoT in prediction of faults and its timing were not considered, while designing the solar based BLDC Motor pumps [13]. Use of synchronous reluctance motor in PV based motor pumping system, proposed by [14] is a recommended choice for energy management but the smartness lacks due to non-availability of IoT usage. The study carried out to compare the effectiveness of reluctance and BLDC Motors in off grid pump systems, have not considered the effect of IoT in increasing the durability and trust [15].

Introduction of new energy management rules in solar based pumping systems [16] have not addressed the need for fault diagnosis and system monitoring using IoT. Though [17] presents the distinct features of PV, wind and hybrid pump systems, the inclusion of IoT for failure management is not carried out. Though various novel methodologies were introduced in the pumping system like reluctance motors with reduced number of poles [18] and three phase based PV pumping system [19], the focus on improving the system life and durability using IoT is still missing. Solar pumping systems with MPPT driven by smart hysteresis rules [20] failed to bring in the advantage of IoT in failure monitoring and prevention. Design of slide mode control [21] and joint power interlinked solar water pumping system [22] doesn't possess the necessary mechanism to do repair management. Though the solar pumping system with Pulse Width Modulation (PWM) control rules [23] and IoT based PV weed cutting machines [24] are noteworthy IoT dependent systems, deeper analysis as regards the system care is missing. The review works detailed by [25] highlights only the energy saving technologies but not about the system caring using IoT tools. The Model Predictive Control (MPC) rules framed for PV based pump operation [26] and proposition of novel coil mixed motors [27] were not completely smart, as they haven't included

any IoT based data tools. Smart inverters recommended for sensor less operation of Permanent Magnet BLDC (PMBLDC) motors [28] also failed to include IoT tools for diagnostic tracking.

Various technologies like watering of plants via learning models, energy efficient pumping system [29], predictive speed systems [30] all lack real time observation and safety setups. The smart PV Controlled system reported in [31] as well as PR rules-based system suggested by [32] have not included any components of IoT. In [33], Machine Learning (ML) has been employed to detect the faults, but without IoT, the model couldn't be able to respond quickly. In the same pattern, IoT technique is not preferred in the MPPT drive based on brain model learning [34] and in current disturbances [35]. Though the grid connected reluctance motor based drives [36] and PV based pump drive hardware equipment [37] demonstrate good controlling techniques, real time data monitoring and issue fixing with IoT sensors are lacking. The MPPT control based on brain functioning [38] and the fuzzy logic architectures for smart control [39] have not reaped any benefits of using IoT in system health investigation. Further, there is no inclusion of system health checkup using IoT techniques in the works like hybrid power combination of solar and wind [40], Single-Ended Primary Inductor Converter (SEPIC) based PV pump drives [41] and fuzzy rule based smart MPPT designs [42]. The solar pumping unit proposed by [43] has no centralized rules and IoT based system health check architecture. The same can be observed in PV fed BLDC Motor control using brain modelling [44] and solar energy-based water supply units [45]. Not only these applications, a proper arrangement for monitoring and correcting the pumping system's life and cost are not available even in latest developments like Digital Signal Processor (DSP) based control for PV-BLDC pumps for rural areas [46], Fault analysis study [47] and techno-economic constraint research [48].

The development of modern renewable energy systems now centers on three main areas which include intelligent control systems and efficient power conversion methods and data-driven modeling techniques. The deep learning (DL)-based PV power forecasting methods research study examined various advanced architectural models which included deep transfer learning and federated learning and probabilistic forecasting models and hybrid DL frameworks to determine their impact on prediction accuracy and scalability [49]. The renewable energy systems developed ML control strategies which used an Improved Recurrent Radial Basis Function Neural Network (IRRBFFNN) system that utilized Modified Particle Swarm Optimization (MPSO) for wind turbine pitch control under dynamic conditions [50]. The research in data-driven modeling approaches used linear regression and Support Vector Machines (SVM) and Decision Trees (DT) and Random Forest (RF) and transfer learning methods to predict performance and analyze degradation in proton exchange membrane fuel cells (PEMFCs) [51]. The implementation of state-space modeling techniques enables the analysis and simulation of DC-DC converters which include Boost and Buck and Buck-Boost topologies to assess system design elements and performance metrics [52, 53]. The research investigates how different voltage boosting methods which include switched inductor and switched capacitor and voltage multiplier and multistage converter topologies can enhance power conversion efficiency during different operational conditions. The research demonstrates how modern renewable energy systems depend on intelligent algorithms together with advanced converter designs yet their development remains restricted when combined with IoT-based systems for real-time monitoring and predictive maintenance of PV-fed BLDC water pumping systems.

Though there is a steep rise in the adoption of solar powered BLDC water pumping systems, the deficit in instant tracking and error spotting has led to multiple issues like defective functioning, abrupt breakdowns and unexpected system halting. These are directly linked with the economics of the system operation. Especially, the delayed identification of faults and maintenance may cause increased operating and repairing expenses. Further, without the use of IoT and ML for system monitoring, it is highly difficult in managing the dynamic weather conditions. This is because, without proper means for detecting and alarming the uncertainties, regular work management will be difficult leaving to a hard time. Thus, it is vital to bridge the missing gap so that the overall trust of the system is increased with reduced costs and improved lifetime. To address these concerns, this research work proposes an IoT based real time monitoring and maintenance prediction system suitable for solar powered BLDC water pumping units. The main objective is to set up a coalition system that uses ML for monitoring the key parameters, detecting the faults at an early stage, checking the system functioning using real time test cases and

finally displaying the results and warning using cloud-based data. All these in turn contribute to increased operating speed, credibility and durability of the water pumping system. Further, this added feature will also convert the system into smarter and preplanned units. The work proposed in this report contains a strong connection between the IoT based arrangement and the cloud stored data. Possibility of failures is predicted in advance by the ML tools with greater accuracy and the failure corrections are being done using fuzzification rules. Thus, the arrangement provides unlimited monitoring feature offered by IoT and the data operation controlled by cloud server. The results obtained clearly highlight the cost reduction and smart energy management in the water pumping systems.

2 Proposed Framework for IoT-Enabled Monitoring and Predictive Control

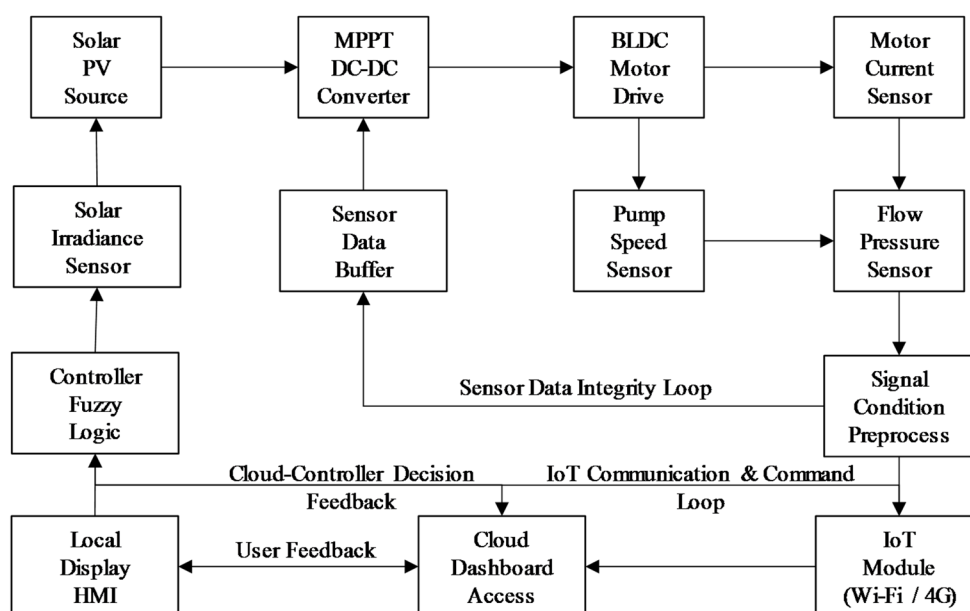
The proposed IoT based architecture has been discussed in this section. The system architecture brings together both the electrical features and the water flow parameters, as it is required to maintain steady water flow and stable power quality. In that sense, various sensors, cloud data storage and control loops are incorporated into the system. Earlier, actions were taken based on the conditions. But this system helps in energy control even under varying sunlight intensities. Moreover, as the motor and pump operation are controlled by data lines, the backup feature comes into action whenever there is a delay in data reception or disturbances in signals. Following this section, the PV based BLDC motor system is presented along with their controlling techniques in a step-by-step fashion.

2.1 Integrated Architecture of PV-Fed BLDC Water Pumping System

Figure 1 demonstrates the various blocks involved in the proposed system architecture. It can be seen that the Solar PV Source is interconnected with BLDC Motor through MPPT controlled converters. To detect the sunlight, current, speed and water flow rate, appropriate sensors are used.

All the sensors employed ensure the collection of data at all instances. Adjustment of signals and early-stage monitoring will help in better data control. The modified data after error correction is sent to the control block after due storing. The control logic used here is of fuzzy logic type. Simultaneously, the data is stored in the cloud using IoT. This cloud linkage will help in monitoring and decision making. Based on the processed data from

Fig. 1 System architecture of the IoT-enabled PV-fed BLDC water pumping system with integrated sensing, control, and cloud communication



cloud, the controller is activated for necessary steps. Thus, the cloud console confirms timely diagnosis of system through supervision from remote distances. Further, a separate block is also allotted to store the data values locally.

2.2 Modeling of PV

The PV module converts solar irradiation into electrical energy through semiconductor junctions. The electrical behavior of a PV cell can be represented using a single-diode equivalent circuit consisting of a current source, diode, series resistance, and shunt resistance [54]. This model accurately describes the nonlinear current-voltage characteristics of the PV array under varying temperature and irradiance conditions.

The output current of the PV array is expressed as

$$i = n_p \left[i_{PV} - i_{rs} \left(e^{\frac{Q(v + ir_{se})}{\alpha n_{se} k t}} - 1 \right) \right] - \frac{v + ir_{se}}{r_{sh}} \quad (1)$$

where i indicates output current of the PV array, n_p denotes number of parallel-connected PV strings, i_{PV} refers photocurrent generated by the PV module, i_{rs} represents reverse saturation current of the diode, Q indicates electron charge, v denotes output voltage of the PV module, r_{se} refers series resistance, r_{sh} denotes shunt resistance, n represents diode ideality factor, k indicates boltzmann constant, and t refers cell temperature. The photocurrent generated by the PV module depends on solar irradiance and temperature and can be calculated as:

$$i_{pv} = [i_{sc} + k(t - t_{Ref})] \frac{s}{s_{Ref}} \quad (2)$$

where i_{sc} refers short circuit current, t_{Ref} denotes reference temperature, S represents solar irradiance, and s_{Ref} indicates reference irradiance.

The diode reverse saturation current can be expressed as:

$$i_{rs} = \frac{i_{sc} + k(t - t_{Ref})}{e^{\left[\frac{Q[v_{ov} + k_v(t - t_{Ref})]}{n_{se} a k t} \right]} - 1} \quad (3)$$

where v_{ov} denotes open circuit voltage, indicates temperature coefficient of open circuit voltage, n_{se} represents number of series-connected PV cells.

Generally, the PV array connected to a DC-DC boost converter and MPPT controller to achieve maximum power output during different environmental conditions.

2.3 Modeling of BLDC Motor

The PV irrigation systems use BLDC motors because they provide high efficiency and dependable operation with low maintenance needs. The BLDC motor operates as a three-phase synchronous motor that uses permanent magnets for its rotor design and three-phase windings for its stator construction. The phase voltage equations describe the electrical behavior of the BLDC motor through their representation of stator resistance and inductance and back electromotive force (EMF) components. The equations shows how phase voltages and currents combine to create back EMF which is generated in the system.

The phase voltage equations of the BLDC motor are expressed as:

$$v_a = l \frac{di_a}{dt} + Ri_a + e_a \quad (4)$$

$$v_b = l \frac{di_b}{dt} + Ri_b + e_b \quad (5)$$

$$v_c = l \frac{di_c}{dt} + Ri_c + e_c \quad (6)$$

where v_a , v_b , and v_c represents phase voltages of the BLDC motor, i_a , i_b , and i_c are phase currents, R denotes phase resistance, l indicates phase inductance, and are back electromotive force of each phase.

The above equations can be represented in matrix form as:

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = R \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} l & 0 & 0 \\ 0 & l & 0 \\ 0 & 0 & l \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (7)$$

Assuming that the three-phase windings are symmetrical and there is no mutual inductance between the phases, the inductances are equal:

$$l_a = l_b = l_c = l \quad (8)$$

The back electromotive force (Back EMF) generated in each phase depends on the rotor position and angular velocity. The back EMF of phase A is given by:

$$e_a = k_e f(\theta_e) w_q \quad (9)$$

Similarly, the back EMF of b and c are expressed as:

$$e_b = k_e f\left(\theta_e - \frac{2\pi}{3}\right) w_q \quad (10)$$

$$e_c = k_e f\left(\theta_e + \frac{2\pi}{3}\right) w_q \quad (11)$$

where k_e indicates the motor back EMF constant, w_q represents rotor angular velocity and θ_e denotes electrical position rotor angle.

The electromagnetic power produced by the BLDC motor is expressed as:

$$p = t_e w_q \quad (12)$$

where P denotes output mechanical power, t_e refers electromagnetic torque.

2.4 Modeling of Water Pump

The hydraulic load torque of the pump is proportional to the square of the flow rate

$$T_L = K_h \cdot Q^2 \quad (13)$$

Where the hydraulic coefficient is indicated as K_h and water flow rate as Q .

The volumetric flow rate is given by,

$$Q = A \cdot v = A \cdot \sqrt{\frac{2(P_{out} - P_{in})}{\rho}} \quad (14)$$

Where the pipe cross sectional area is denoted as A , water velocity as V , outlet and inlet pressures as P_{out} and P_{in} , and the water density as ρ .

The pump's hydraulic power is

$$P_{hyd} = \rho \cdot g \cdot Q \cdot H \quad (15)$$

Where the gravitational acceleration is represented as g and the pump head as H . Equation (15) demonstrates the active work carried out by the pump.

2.5 Objective Function

The overall system efficiency is expressed as

$$\eta_{sys} = \frac{P_{hyd}}{P_{pv}} \cdot 100\% \quad (16)$$

Where η_{sys} symbolizes the total system efficiency. This factor signifies the amount of conversion of solar energy into mechanical pumping energy. Deficit in efficiency if any demands suitable predictive maintenance plans.

The Total Harmonic Distortion (THD) for current monitoring is

$$THD = \frac{\sqrt{\sum_{n=2}^{\infty} I_n^2}}{I_1} \quad (17)$$

where the fundamental current is given by I_1 and harmonic components by I_n .

2.6 Deployment of Sensor Nodes and Real-Time Data Acquisition Pipeline

The IoT-based monitoring system requires both sensor node deployment and real-time data acquisition system development to achieve its goal of monitoring and predictive maintenance of the PV-fed BLDC irrigation pumping system. The data flow architecture is illustrated in Fig. 2, which shows the interaction between sensing units, the edge controller, the IoT gateway, and the cloud monitoring platform.

The PV-fed BLDC irrigation pumping system requires multiple sensors to track their electrical mechanical and hydraulic system parameter which ensures precise system monitoring and dependable system operation. On the PV side, a voltage sensor is used to measure the PV array output voltage with a measurement range of 0–300 V and an accuracy of $\pm 0.5\%$, while a Hall-effect current sensor measures the PV current within a range of 0–20 A with an accuracy of $\pm 1\%$. The operating condition of the BLDC motor is monitored using a Hall-effect speed sensor capable of measuring rotational speeds up to 5000 rpm with a resolution of 1 rpm. A digital temperature sensor with an operating range of -40 °C to 125 °C and an accuracy of ± 0.5 °C is installed near the motor housing to monitor thermal variations and prevent overheating. For hydraulic performance monitoring, a turbine-type flow sensor is used to measure the water discharge rate within a range of 0.3–6 m³/h with an accuracy of $\pm 2\%$, while a pressure sensor measures pipeline pressure within a range of 0–10 bar with an accuracy of $\pm 1\%$. The motor housing of the system contains MEMS accelerometer which can measure vibrations between ± 16 g to detect mechanical faults including bearing wear and rotor imbalance. The edge controller samples all sensor signals every second, processes them, and sends the data to the IoT gateway. There, the data is stored in the cloud for monitoring purposes and predictive maintenance analysis. Sensor specifications used in the proposed system are tabulated in Table 1.

When solar PV power becomes available, the system initialization process activates the sensing and communication modules. The initialization process requires all sensors which connected to the solar array and BLDC

Fig. 2 Flowchart illustrating the real-time data acquisition process from solar, motor, and hydraulic sensors to the IoT cloud platform

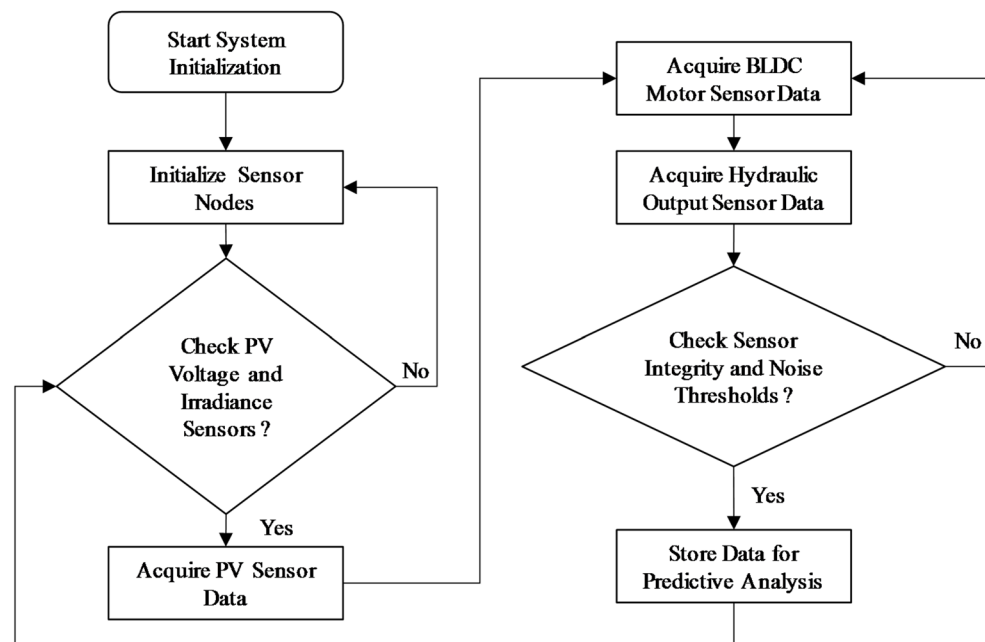


Table 1 Sensor specifications used in the proposed IoT-enabled irrigation monitoring system

Sensor Type	Measured Parameter	Measurement Range	Accuracy	Resolution
Voltage Sensor	PV array voltage	0–300 V	±0.5%	0.1 V
Hall-effect Current Sensor	PV current	0–20 A	±1%	0.01 A
Hall-effect Speed Sensor	Motor speed	0–5000 rpm	±1 rpm	1 rpm
Digital Temperature Sensor	Motor temperature	–40–125 °C	±0.5 °C	0.1 °C
Turbine Flow Sensor	Water flow rate	0.3–6 m ³ /h	±2%	0.01 m ³ /h
Pressure Sensor	Pipeline pressure	0–10 bar	±1%	0.01 bar
MEMS Accelerometer	Motor vibration	±16 g	±5%	0.01 g

motor and hydraulic pump to undergo calibration while establishing reference values for their measurements. The system conducts automatic reinitialization procedures when it identifies any sensor malfunction because this process ensures continuous system reliability. Once the system is initialized, the control unit verifies whether the PV voltage and solar irradiance are within the predefined operational limits required for safe motor operation. If these conditions are not satisfied, the BLDC motor is prevented from starting in order to avoid damage caused by insufficient input power. The system starts collecting data when the required operating conditions are met. The sensor nodes which have been installed throughout the irrigation system continuously monitor several electrical and hydraulic system measurements. The PV system uses sensors to gather data about PV output voltage and PV current and module temperature. These measurements provide information about solar energy availability and PV system health. The motor side sensors monitor essential operational data which includes phase currents and motor speed and electromagnetic torque to assess BLDC motor efficiency. The irrigation system performance assessment at the pumping unit receives data from hydraulic sensors which monitor irrigation system efficiency. The sensors compute water flow rate and pipeline pressure and discharge head. The system uses these parameters to evaluate hydraulic efficiency while also detecting abnormal conditions which include pipe blockage and pump degradation. The edge controller collects all sensor data which it processes by assigning timestamps to each data packet before sending it out. The system checks the packets for accuracy and completeness before sending them to the IoT gateway through the communication network. The gateway system requires incoming data packets to undergo two verification tests which check for data integrity and synchronization accuracy. The cloud storage platform receives validated data which the platform uses to store and analyze data while displaying results

through the IoT dashboard. The cloud interface enables remote monitoring of system parameters which include PV output power and motor operating conditions and irrigation discharge rate. The edge controller executes preprocessing tasks on sensor information which it receives from the PV array and BLDC motor and hydraulic pumping unit before sending the information to the cloud platform. Initially, noise filtering is applied to remove transient measurement disturbances using a moving average filtering technique. The dataset maintains temporal continuity through linear interpolation which fills in missing data samples that resulted from communication interruptions. The system uses min-max scaling to convert all PV voltage and PV current and motor current and motor speed and temperature and water flow rate measurements into standardized numerical ranges for ML analysis. The system identifies and removes all abnormal sensor readings which occur due to temporary measurement errors through statistical outlier detection methods. The collected data reliability improves through these preprocessing steps which also boost the predictive maintenance model's ML prediction performance. The process for data acquisition and communication operates without interruption to enable continuous real-time monitoring and predictive maintenance and system performance assessment. The dataset generated through this monitoring pipeline supports both ML-based fault prediction model training and validation processes which are described in the subsequent sections.

3 Logistic Regression Model for Predictive Maintenance and IoT Monitoring

The proposed IoT-enabled monitoring framework uses a supervised ML method to support predictive maintenance of the PV-fed BLDC irrigation system. The system control uses Fuzzy Logic controller as its main control system while the system requires fault prediction and predictive maintenance through a ML model which utilizes Logistic Regression. The logistic regression classifier uses real-time sensor measurements from the PV array and BLDC motor and hydraulic pumping unit to predict potential system faults. Logistic regression is selected due to its computational efficiency, interpretability, and suitability for real-time embedded monitoring systems. The model functions as a binary classification system which calculates fault probabilities using PV voltage and PV current and motor phase current and motor speed and motor temperature and vibration level and water flow rate and discharge pressure as input variables. The model uses historical telemetry data from the IoT monitoring platform to train its system which contains two labeled operational states of normal operation and fault condition. The trained classifier continuously examines incoming sensor data while it calculates the likelihood of unusual system behavior. The system activates a maintenance alert through the IoT dashboard when the predicted probability reaches the predefined threshold which enables early fault detection and supports preventive maintenance activities.

To enable predictive fault detection in the proposed PV-fed BLDC irrigation system, a supervised ML model based on logistic regression was implemented. The model was trained using historical telemetry data collected from the IoT-enabled monitoring framework. The dataset contains time-stamped sensor measurements which were obtained from the PV array and BLDC motor and hydraulic pumping unit. The system monitored eight parameters which included PV voltage and PV current and motor phase current and motor speed and motor temperature and vibration level and water flow rate and discharge pressure. Each observation in the dataset was labeled either as normal operation (0) or fault condition (1) based on historical operational logs and maintenance records. Prior to training the ML model, the acquired sensor data were subjected to preprocessing procedures to improve model reliability and data consistency. Missing values caused by communication interruptions were handled using linear interpolation, while all input features were normalized using min-max scaling to ensure consistent numerical ranges across the dataset. In addition, abnormal measurements resulting from temporary sensor noise were removed using a statistical outlier detection technique. The complete dataset was then divided into training and validation subsets using a 70:30 split ratio, where the training dataset was used to estimate the model parameters and the validation dataset was used to evaluate the prediction performance and generalization capability of the model. The logistic regression classifier was trained using the maximum likelihood estimation

method, in which the model parameters were optimized through a gradient descent algorithm. During training, the learning rate was set to 0.01 with a maximum of 500 iterations, and L2 regularization with a coefficient of 0.1 was applied to prevent overfitting and improve model stability. The model which was developed uses real-time sensor data from the IoT monitoring system to calculate fault probabilities. When the predicted probability exceeds a predefined threshold, the system identifies the condition as a potential fault and triggers a maintenance alert through the IoT dashboard. The classification model was tested with standard performance metrics which included accuracy and precision and recall and F1-score to show that it can effectively detect faults and support predictive maintenance in the PV-fed BLDC irrigation pumping system.

3.1 Logistic Regression Fault Prediction Model

The predictive maintenance system uses a logistic regression classifier which calculates the probability of faults based on real-time sensor data from the PV system and BLDC motor and hydraulic pump. The embedded monitoring systems use logistic regression because it provides a simple and interpretable method to solve binary classification tasks. The probability of fault occurrence is expressed as

$$P_{fault} = \frac{1}{1 + e^{-(\beta_0 + \sum \beta_i S_i)}} \quad (18)$$

where fault probability is modelled as P_{fault} , regression weights as β_i and sensor inputs as S_i .

A maintenance alarm is triggered when the predicted probability exceeds a predefined threshold.

$$P_{fault} > \delta \quad (19)$$

Where the user defined threshold is taken as δ . The maintenance alarms are displayed on the IoT dashboard, allowing early identification of system faults before catastrophic failure occurs.

3.2 IoT-Based Data Acquisition and Transmission Model

To develop a real-time data monitoring architecture suitable for PV powered BLDC irrigation pumping system, it is essential to model the flow of data acquired from various sensors placed on solar cells, electrical motors and hydraulic flow meters to the cloud framework. The total data collected from all the sensors necessary for cloud synchronization is best described as

$$R_{cloud} = \sum_{i=1}^n (f_{s_i} \cdot b_{s_i} \cdot \gamma_i \cdot \delta_i) \quad (20)$$

Where the total data rate is represented as R_{cloud} in bits per second, sampling frequency of the i^{th} sensor as f_{s_i} (Hz), number of bits per sample as b_{s_i} , compression efficiency factor as γ_i and the transmission overhead factor as δ_i . The amount of impact created on cloud data ingestion rate due to sensor nature and communication protocol overheads are quantified by this equation. With high value of R_{cloud} , it is necessary to introduce the real-time feedback delays, as the demand on the IoT channel will be huge.

To analyze the communication delay between the edge controller and the IoT cloud platform, the transmission latency is modeled as

$$D_{tx} = \frac{\sum_{j=1}^m S_j \cdot (1 + \eta_j)}{B_{eff} \cdot (1 - \rho)} + \tau_q \quad (21)$$

Here, the total transmission delay is symbolized as D_{tx} seconds, data size of source ‘j’ as S_j , j^{th} stream’s protocol overhead ratio as η_j , effective bandwidth of the communication link as B_{eff} , channel utilization factor as ρ and queuing delay at the gateway as τ_q . This equation replicates the real time behavior of the system, when affected by the transmission impediments and data attributes. To effectively monitor the system data like solar PV voltage and pump output, visually on the cloud dashboard, this D_{tx} value needs to be maintained within the acceptable range.

3.3 Real-Time Monitoring Index

A normalized monitoring index is established, which shows the complete operational state of the PV irrigation system. This index combines multiple sensor measurements, which include PV voltage and motor current and water discharge rate.

$$\Lambda(t) = \frac{1}{3} \left(\frac{V_{pv}(t) - V_{ref}}{V_{ref}} + \frac{I_{bldc}(t) - I_{nom}}{I_{nom}} + \frac{Q_{water}(t) - Q_{min}}{Q_{max} - Q_{min}} \right) \quad (22)$$

where PV voltage at a time t is given by $V_{pv}(t)$, motor phase current by $I_{bldc}(t)$, volumetric water discharge rate by $Q_{water}(t)$, and the reference values used are V_{ref} , I_{nom} , Q_{min} , Q_{max} . Thus, the operational efficiency can be visually observed using this normalized dynamic index value, $\Lambda(t)$, that is directly proportional to the system stress. Higher values of this index indicate increased system stress and trigger predictive fault diagnosis.

3.4 Predictive Maintenance Optimization

The logistic regression model estimates the probability of fault occurrence using historical fault-labeled telemetry and real-time sensor measurements.

$$P_{fault}(t) = \frac{1}{1 + e^{\left(- \left(\theta_0 + \sum_{k=1}^n \theta_k x_k(t) \right) \right)}} \quad (23)$$

where real-time parameters like temperature of the inverter, pressure sensor drift, rotor current are denoted as $x_k(t)$, and the trained logistic regression coefficients as θ_k .

To minimize maintenance costs and system downtime, the optimal maintenance interval is determined using cost-risk optimization.

$$T_{opt} = \min_T \left(C_m \cdot \lambda_f(T) + C_d \cdot \int_0^T D(t) dt \right) \quad (24)$$

Where the maintenance cost is denoted as C_m , time-dependent failure rate as $\lambda_f(T)$, cost of degradation-caused performance loss as C_d , and degradation function over a time as $D(t)$. This cost-risk based optimization function provides an efficient service routine. With proper tuning of T_{opt} , along with improvement of motor life, the possibility of severe failures can also be prevented.

3.5 Dynamic BLDC Motor Torque Model

The electromagnetic torque of the BLDC motor under varying solar irradiance conditions is expressed using the d-q axis dynamic model.

$$T_e(t) = \frac{3}{2} \cdot \frac{P}{2} \cdot (\psi_d(t) \cdot i_q(t) - \psi_q(t) \cdot i_d(t)) \quad (25)$$

where $T_e(t)$ denotes instantaneous torque output, P represents number of motor poles, $\psi_d(t)$ and $\psi_q(t)$ indicates the d-axis and q-axis flux linkages respectively, and $i_d(t)$, $i_q(t)$ are the d-axis and q-axis currents got from Clarke-Park transformation. Variations in PV output directly influence motor torque and consequently the hydraulic pumping performance.

3.6 System Efficiency Analysis

The instantaneous system efficiency is calculated using the ratio between hydraulic output power and PV electrical input power.

$$\eta_{sys}(t) = \frac{\rho \cdot g \cdot h(t) \cdot Q(t)}{V_{pv}(t) \cdot I_{pv}(t)} \cdot 100\% \quad (26)$$

Where water density is symbolized as ρ , gravitational acceleration as g , pumping head as $h(t)$, water flow rate as $Q(t)$ and PV Panel voltage and current as $V_{pv}(t)$ and $I_{pv}(t)$ respectively.

A performance indicator based on utilization of energy and signal latency is defined as

$$\Xi = \frac{\int_0^T P_{use}(t) dt}{\int_0^T P_{gen}(t) dt} \cdot \left(1 - \frac{\bar{D}_{IoT}}{\bar{D}_{legacy}} \right) \quad (27)$$

where the power consumed by the load is represented as $P_{use}(t)$ and generated PV power as $P_{gen}(t)$. While $\bar{D}_{IoT}$ indicate the average end to end delays in an IoT system, the $\bar{D}_{legacy}$ refers to that of traditional system. To obtain an optimized efficiency and enhanced responsiveness, the value of Ξ should be higher.

3.7 System Stability and Reliability Analysis

A Lyapunov function is used to evaluate the stability of the BLDC motor system under predictive control.

$$V(t) = \frac{1}{2} J \omega^2(t) + \frac{1}{2} L (i_d^2(t) + i_q^2(t)) + \frac{1}{2} K_e^2 \theta_e^2(t) \quad (28)$$

Here, the rotor inertia is denoted as J , the motor angular velocity as $\omega(t)$, stator inductance as L , Back EMF constant as K_e , and the electrical position error as $\theta_e(t)$. Variation of $V(t)$ value with respect to time is a check of system stability when subjected to forecasted disturbance.

The relationship between hydraulic flow rate and motor operating parameters is modeled using a regression Eq.

$$Q(t) = \beta_0 + \beta_1 \cdot T_e(t) + \beta_2 \cdot \omega(t) + \beta_3 \cdot I_q(t) + \varepsilon(t) \quad (29)$$

Here, the water flow rate is represented as $Q(t)$, the motor torque as $T_e(t)$, speed as $\omega(t)$, quadrature current as $I_q(t)$, and the model error term as $\varepsilon(t)$.

The Mean Time To Failure (MTTF) is calculated to determine the reliability of the IoT supervised system.

$$MTTF = \frac{\sum_{i=1}^n T_{uptime}^{(i)}}{n} \quad (30)$$

Where the uptime before the i^{th} failure is denoted as $T_{uptime}^{(i)}$ and the number of observed fault instances as n . The system reliability increases with higher MTTF values because the predictive maintenance method shows greater effectiveness.

4 Results

This section describes the simulation carried out to examine the proposed control system post the data acquisition pipelining. The various inputs received from the sensors are fed into the system model.

4.1 Simulation setup and system Configuration for System Implementation

The inputs to the system model are the data packets that are verified in their first stage. Thus, the pitch behavior is properly structured by the simulation arrangement. The simulation parameters considered and their values are listed in the Table 2.

A PV array rated at 1.5 kW is employed to supply power to the 1.3 kW BLDC motor driving the water pump. The PV system capacity is selected to closely match the motor load while accounting for conversion losses in

Table 2 Simulation parameters for the PV-fed BLDC motor and water pumping system

Component	Parameter	Value	Unit
PV Array	PV Module Model	BMU/214	-
	Rated Power of PV Array	1.5	kW
	Max Power Point Voltage (V_{mpp})	28.5	V
	Max Power Point Current (I_{mpp})	7.5	A
	Number of Modules (Series $\times$ Parallel)	6×1	-
	Irradiance at STC	1000	W/m ²
	Cell Temperature at STC	25	°C
DC-DC Converter	Air Mass (AM)	1.5	-
	Converter Type	MPPT-enabled Boost Converter	-
	Switching Frequency	20	kHz
BLDC Motor	Duty Cycle Range	0.2–0.8	-
	Rated Power	1.3	kW
	Rated Speed	3000	rpm
	Rated Voltage	270	V (DC)
	Rated Current	4.8	A
	Rotor Inertia	0.0008	kg·m ²
	Back EMF Constant	0.12	V·s/rad
	Stator Resistance	1.2	Ω
	Stator Inductance	4.5	mH
	Discharge Rate	1.5	m ³ /h
Water Pump	Total Dynamic Head	30	m
	Pipe Diameter	25	mm
	Pump Efficiency	72	%
	Grid Support Voltage	180	V (AC)
Power Supply for Test	Grid Frequency	50	Hz
	Sensor Types	Voltage, Current, Flow, Pressure	-
IoT Infrastructure	Gateway Communication Protocol	MQTT / REST API	-
	Cloud Platform	AWS / Azure	-
	Data Sampling Interval	1	s
	Edge Device Preprocessing	Enabled (filtering, normalization)	-
Simulation Platform	Modeling Tool	MATLAB Simulink	-
	Time Step	1×10^{-4}	s
	Simulation Duration	120	s

the DC-DC converter and inverter stages, as well as variations in solar irradiance. This modest design margin (approximately 10–15%) ensures adequate power availability under suboptimal environmental conditions, thereby maintaining stable motor operation and consistent pumping performance. The proposed design achieves better system performance and energy efficiency because it avoids power generation problems and mismatch losses which occur with large system configurations. The MPPT-based boost converter operates by controlling the PV array's operating point which enables it to extract maximum power across different solar irradiance conditions. This configuration ensures proper synchronization between the electrical and mechanical subsystems, enhancing system reliability and operational efficiency across all test scenarios. The proposed system uses optimal PV oversizing because it achieves an ideal balance between energy production and energy requirements.

Figure 3 presents the complete Simulink model of the proposed solar-powered BLDC motor drive system developed for water pumping applications. The system is initiated by a PV array, which converts solar irradiance into electrical energy under varying environmental conditions. The irradiance input is modeled using dynamic profiles to emulate realistic operating scenarios, while the temperature is maintained at standard test conditions. The output voltage and current from the PV array are fed into a MPPT controller, which continuously extracts the maximum available power by generating an optimal duty cycle. This duty cycle is applied to a DC-DC boost converter comprising an inductor, diode, switching device, and output filter components, ensuring voltage regulation and power enhancement. The system first delivers regulated DC output to a three-phase inverter, which transforms the DC into AC that powers the BLDC motor. The BLDC motor operates the centrifugal water pump while the model tracks essential performance metrics which include speed and torque and current measurements.

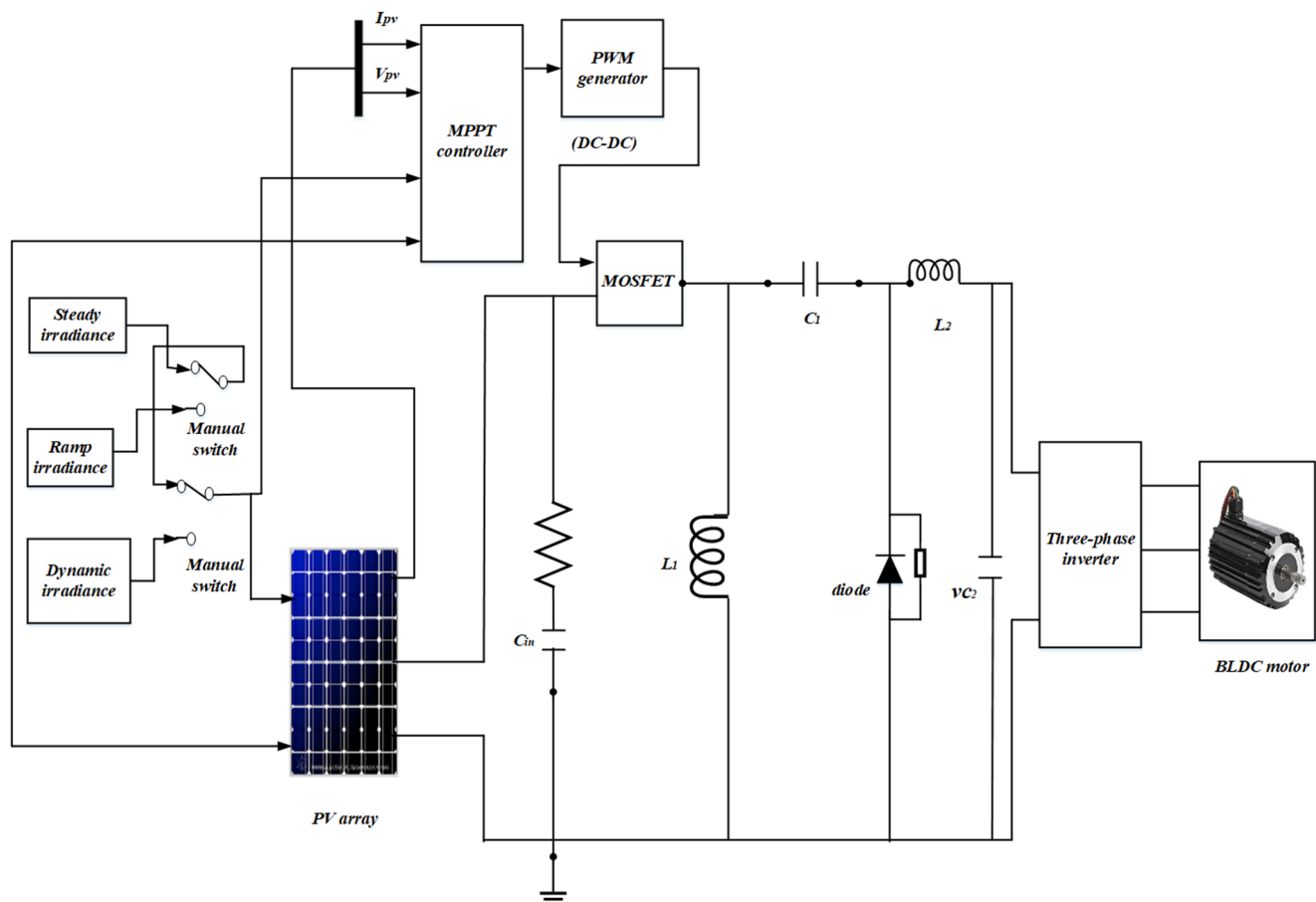


Fig. 3 Simulink-based model of the proposed PV-fed BLDC motor drive system with MPPT and power conversion stages for water pumping

The integrated configuration combines solar energy generation with power conditioning and motor control functions to deliver consistent and efficient pumping performance across different irradiance levels.

4.2 Performance Analysis under Varying Solar Irradiance

Figure 4 illustrates the dynamic variation of solar insolation applied to the system over time. Initially, from 0 to 0.2 s, the irradiance is maintained at a low level of 200 W/m^2 , representing weak solar conditions. At $t=0.2 \text{ s}$, a sudden increase in irradiance to 800 W/m^2 is observed, indicating a transition to high solar availability, which is sustained until approximately 0.4 s. Beyond this point, the irradiance decreases to a moderate level of 600 W/m^2 , where it remains stable until the end of the observation period at 0.6 s. This stepwise variation in solar insolation is intentionally introduced to evaluate the system's dynamic response under changing environmental conditions, enabling analysis of how the PV system, motor, and pumping unit adapt to low, high, and moderate irradiance levels.

Figure 5 illustrates the dynamic variation of electromagnetic torque of the motor-pump system over time. At the initial stage ($t=0 \text{ s}$), the torque starts at approximately 2 Nm, reflecting the startup condition of the BLDC motor. As time progresses, the torque increases rapidly due to acceleration of the motor, reaching around 6–8 Nm within the first 0.05 s. Between 0.05 s and 0.1 s, the torque continues to rise gradually, indicating the transition toward steady-state operation. Beyond 0.1 s, the torque stabilizes in the range of 9–11 Nm, with small oscillations observed due to switching effects and dynamic load interactions. These oscillations represent the inherent electromagnetic and control-induced variations in the system. The proposed system achieves stable torque output which

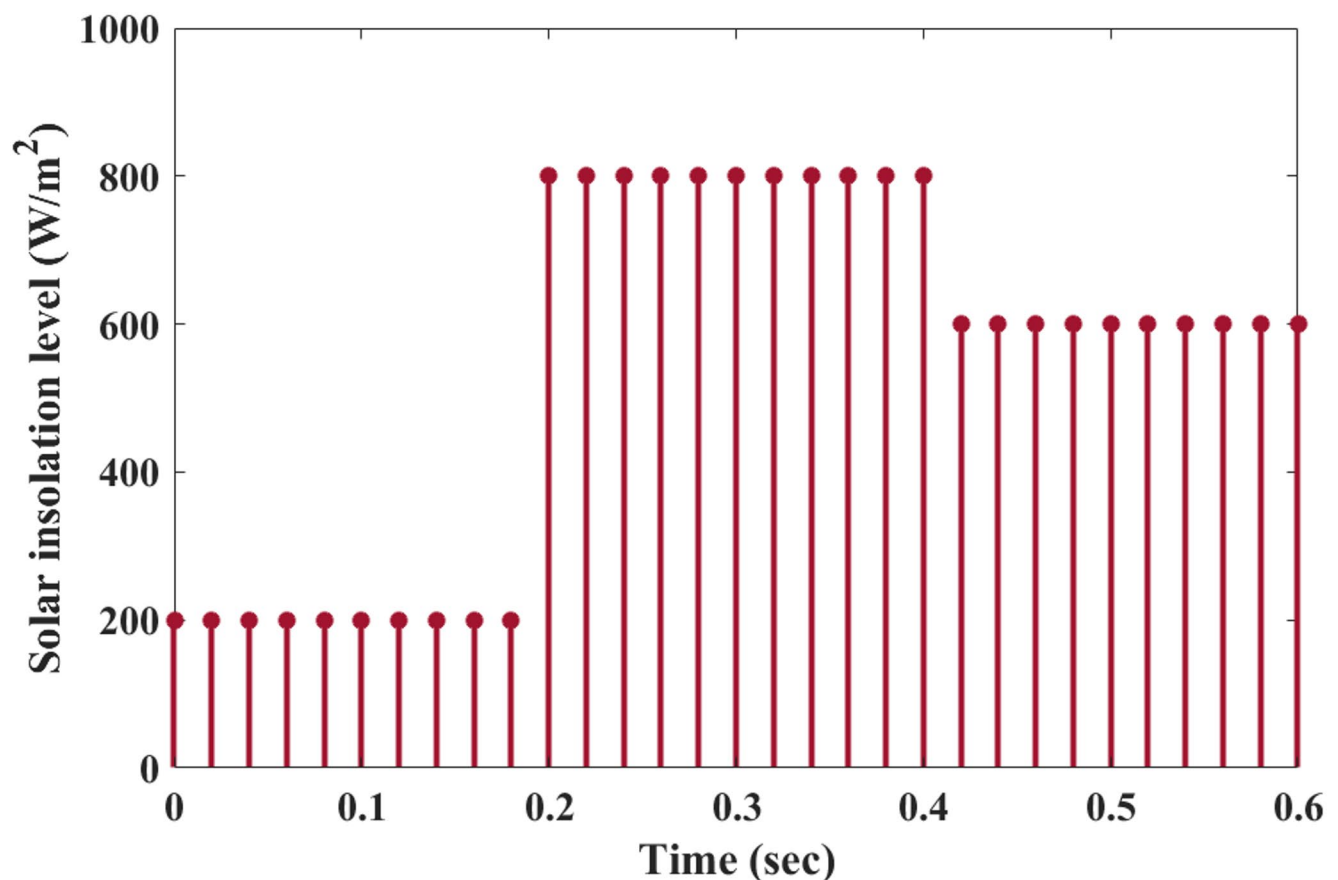
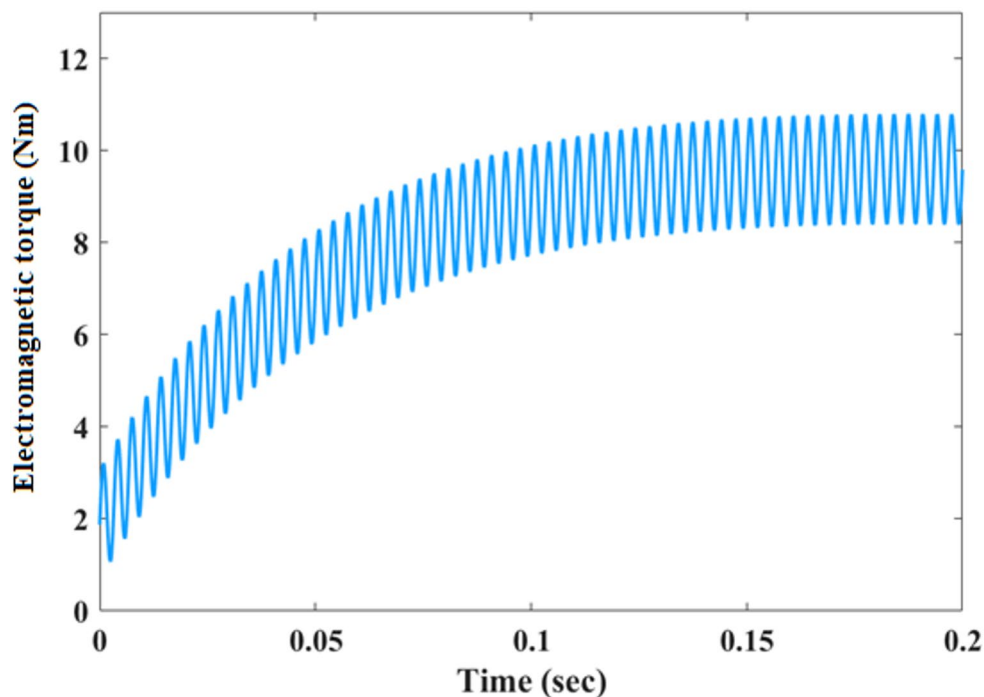


Fig. 4 Analysis of dynamic performance of solar insolation level

Fig. 5 Analysis of motor-pump torque

takes a brief time to reach steady operation while it continues to deliver constant performance during changing operational conditions.

Figure 6 illustrates the dynamic speed response of the BLDC motor driving the pump, where the motor starts from rest at $t=0$ s and accelerates rapidly to approximately 2000 rpm within 0.05 s due to the high initial electromagnetic torque. Between 0.05 s and 0.1 s, the speed continues to increase at a slower rate, reaching around 2700 rpm as the effect of load torque and system inertia becomes significant. Beyond 0.1 s, the motor speed gradually settles to its steady-state value of approximately 2950–3000 rpm by around 0.15 s, with minor oscillations of about ± 20 rpm observed due to switching and commutation effects. The overall response indicates a fast settling time of approximately 0.15 s, negligible overshoot, and stable steady-state operation, demonstrating the effectiveness of the proposed control strategy in ensuring reliable motor performance for pumping applications under dynamic operating conditions.

Figure 7 illustrates the dynamic variation of PV output power under changing solar irradiance conditions. Initially, from $t=0$ s to approximately 0.05 s, the PV power increases rapidly from 0 W to around 500 W, corresponding to low irradiance conditions (200 W/m^2). From 0.05 s to 0.2 s, the power remains nearly constant at approximately 500 W, indicating stable operation under steady low irradiance. At $t=0.2$ s, a sharp increase in PV power to about 2200–2300 W is observed due to the sudden rise in irradiance to 800 W/m^2 , demonstrating effective MPPT. This higher power level is maintained with minimal fluctuation until approximately 0.4 s. Beyond $t=0.4$ s, the PV power drops to around 1600–1700 W as the irradiance decreases to 600 W/m^2 and remains stable thereafter. The smooth transitions and low oscillations (within ± 30 W) indicate fast tracking response, negligible steady-state error, and efficient energy extraction capability of the proposed system under varying environmental conditions.

Figure 8 illustrates the dynamic variation of PV output voltage under changing solar irradiance conditions. Initially, from $t=0$ s to approximately 0.08 s, the PV voltage decreases from about 130 V to around 100 V as the system transitions under low irradiance conditions (200 W/m^2). Between 0.08 s and 0.2 s, the voltage remains relatively stable in the range of 95–105 V with small oscillations (± 5 V). At $t=0.2$ s, corresponding to an increase in irradiance to 800 W/m^2 , the PV voltage rises to approximately 115–120 V and maintains this level with minor ripple. A brief transient dip to near 0 V is observed at around $t=0.4$ s, which can be attributed to switching or

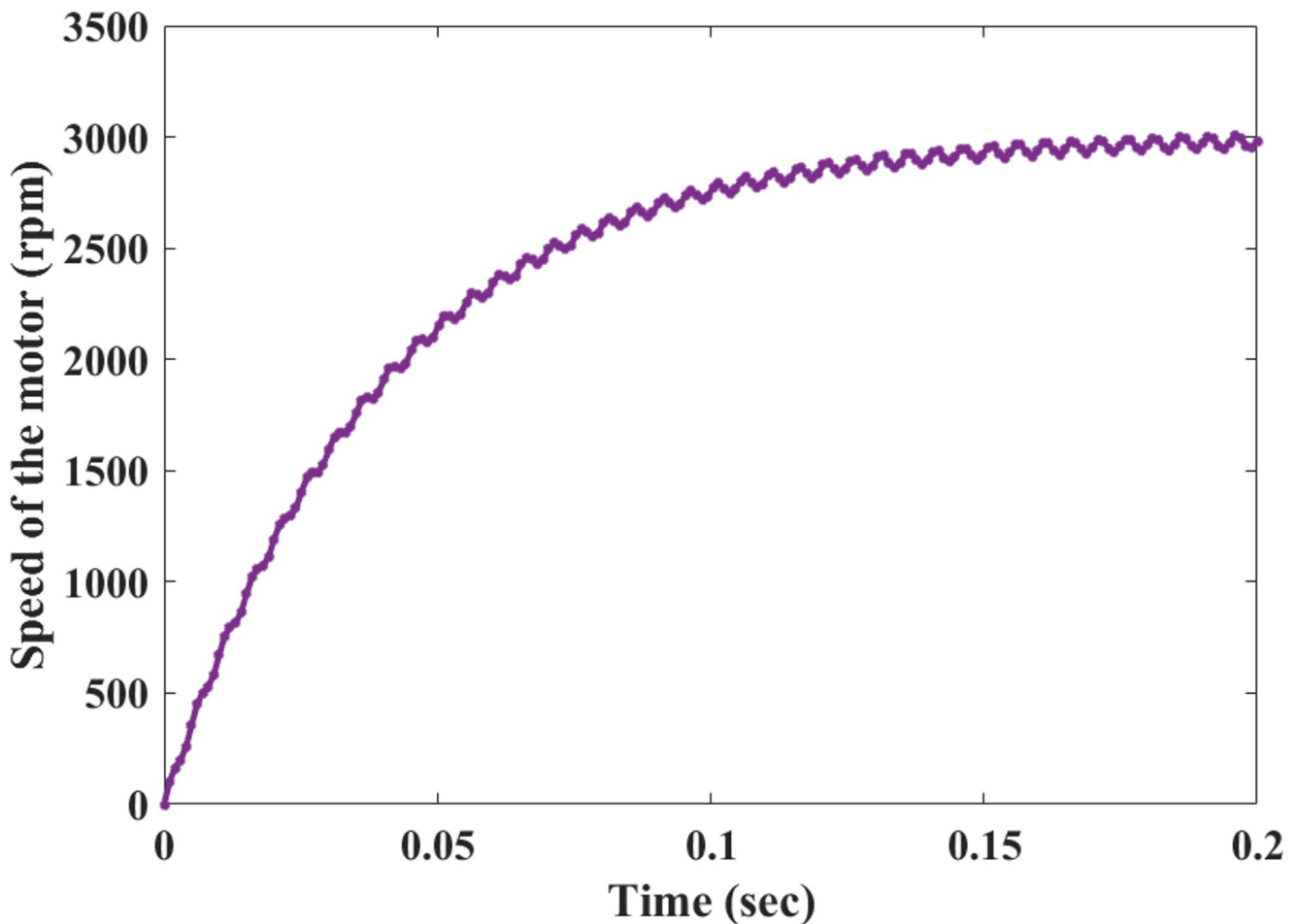


Fig. 6 Analysis of motor-pump speed

control action during the transition to reduced irradiance. Following this disturbance, the voltage quickly recovers and stabilizes again around 110–120 V for the remaining duration (0.4–0.6 s) under moderate irradiance (600 W/m^2). The system shows three essential characteristics which include rapid voltage restoration and small voltage fluctuations and successful PV system control throughout changing environmental conditions.

Figure 9 illustrates the dynamic variation of PV output current under changing solar irradiance conditions. Initially, from $t=0 \text{ s}$ to approximately 0.08 s, the PV current increases gradually from 0 A to about 5–6 A, corresponding to low irradiance conditions (200 W/m^2). Between 0.08 s and 0.2 s, the current remains nearly constant at around 5 A with minor oscillations ($\pm 0.5 \text{ A}$), indicating stable operation. At $t=0.2 \text{ s}$, a sharp rise in current to approximately 20–21 A is observed due to the increase in irradiance to 800 W/m^2 , demonstrating rapid response of the PV system. This higher current level is maintained with small ripple ($\pm 1 \text{ A}$) until approximately 0.4 s. Beyond $t=0.4 \text{ s}$, the current decreases to around 14–15 A as the irradiance reduces to 600 W/m^2 and remains stable thereafter. The fast transitions, low steady-state fluctuations, and proportional relationship with irradiance confirm effective MPPT operation and efficient current regulation under dynamic environmental conditions.

4.3 IoT Cloud Interface for Real-Time Visualization and Remote Monitoring

For effective interfacing with the cloud, the streaming variable data are simplified by the simulation arrangement. This in turn helps in understanding the conditions of PV voltage and motor graphically.

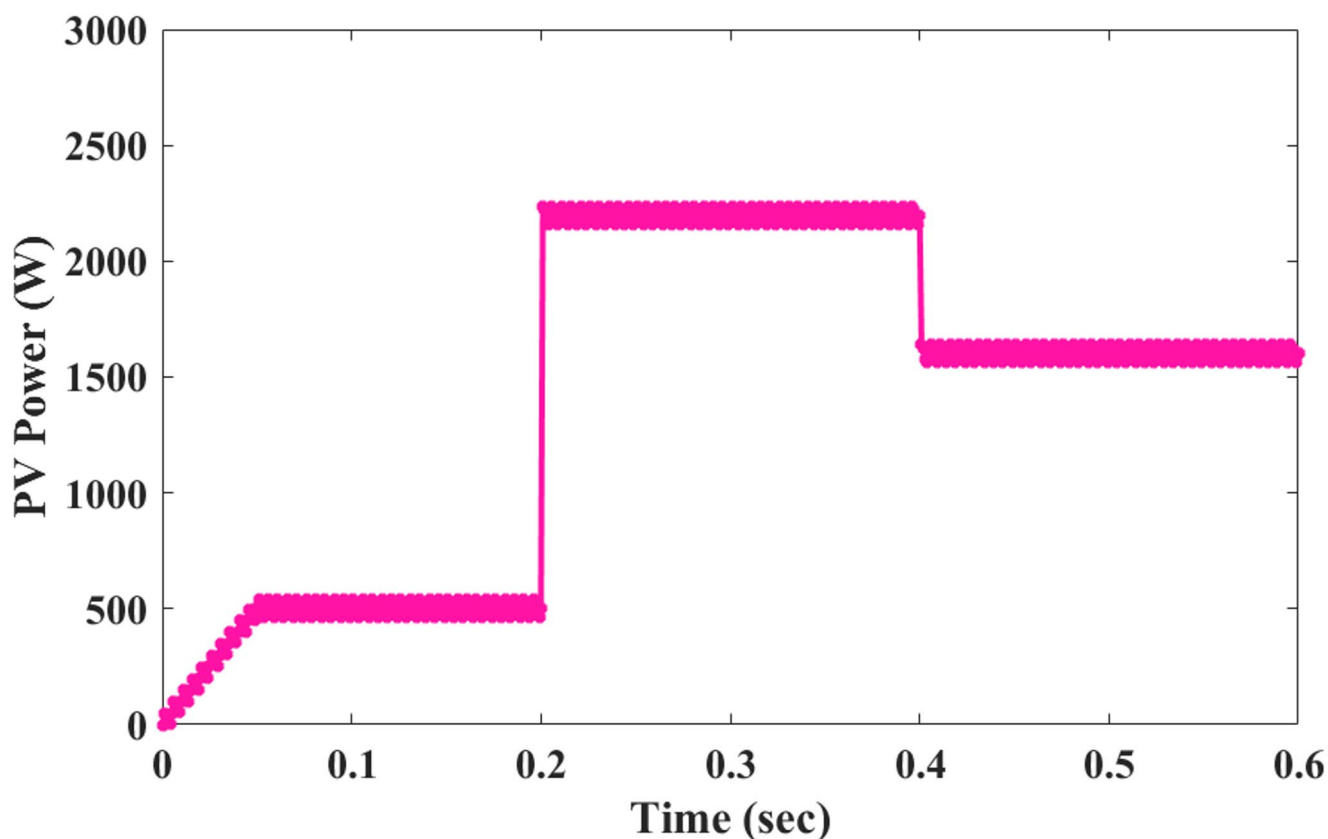


Fig. 7 Analysis of PV power

The variable flow rate values and the electrical signals are displayed in the real time dashboards. It can be observed from Fig. 10 that the relation between the PV voltage and the flow response is non-linear. At above 180 V of PV voltage, there is a steady increase in the discharge rate. But when the voltage falls below 135 V, irrespective of the demand, the discharge rate gets reduced to a greater extent. The variation of motor's torque with variation of irradiance are also considered in these deviations. At the same time, beyond the PV voltage of 210 V, there is an improvement in the current stability. Thus, consistent response is achievable even under load switchovers, which is mainly due to the IoT interface. In case of packet losses, the system initiates retransmission to ensure the data integrity. Readings were adjusted to account for fault detection and to maintain the initiations. Depending on the operational data, the proposed architecture supports instant decision making.

4.4 Predictive Analytics Workflow Using ML Models

For each reading obtained from the sensors, structured data logs are enabled by the cloud interface. The outcomes of the supervised model during fault progression are clearly depicted in Fig. 11. With amplification of temperature and voltage, there is an initiation of performance degradation in the motor. Further, there is a dip in the discharge rate in concurrence with the rise in vibration levels. Categorical classification and regression fits are used by the model for event labeling. At first, the stable input conditions are labeled as normal operations by the model. But as soon as the threshold limit crosses 1.6 L/s and 55 °C, the warning system is activated. Thus, whenever the operational limit is exceeded, the status of fault is updated quickly. Next, the outputs of the supervised model were verified using the recorded downtime data logs. It is observed that vibration levels displayed earlier fault occurrence when compared with the flow metrics. When there is only a temperature rise as against multiple sensor data variations, the fault development seems to be slower.

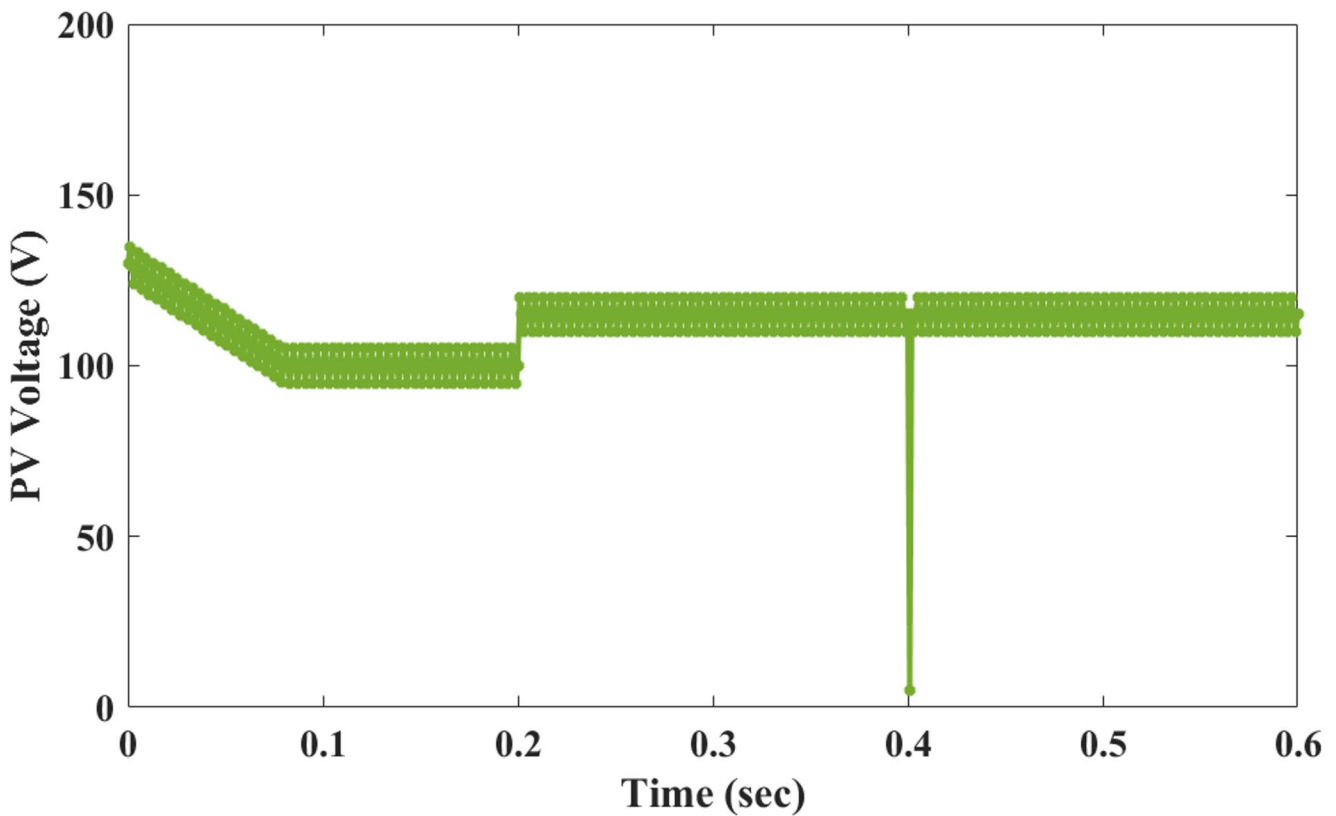


Fig. 8 Analysis of PV voltage

Thus, the merits of combining hydraulic, thermal and dynamic readings altogether are clearly visible in the results. The role of these predictive outcomes in deciding the maintenance timings are discussed in the next section.

4.5 Maintenance Strategy Assessment Based on Predictive Model Output

In continuation to the failure data analysis discussed in the previous subsection, Fig. 12 is demonstrated as a dendrogram, that clusters the anomalies based on their occurrence and progression over a period of time. A sharp rise can be observed during post noon cycles as against the early hours. With the increase in the wear and tear of the system, the possibility of system failure is also high. Thus, categorizing the shorter distances issues helps in reducing the system stresses. Most of the measurements are critically affected by the wear, especially from 10 AM to 6 PM. Further, increased vibration levels signal the development of motor problems every hour. Hence, appropriate measures have to be taken to fix these problems, the failure of which may lead to the halt of the water pumping system. Though the system is recoverable during nighttime, the effect of some amount of stress is still prevalent.

Thus, during the high amount of system wear problems, maintenance schedules should be designed with a specific focus on the timing intervals. As against static intervals, refined planning and control is offered by the predictive thresholds. In short, the scheduling parameters are better optimized by the high-risk clusters.

4.6 Performance Analysis of BLDC Motor under Variable Solar Input

Figure 13 presents the motor performance under fluctuating solar irradiance levels

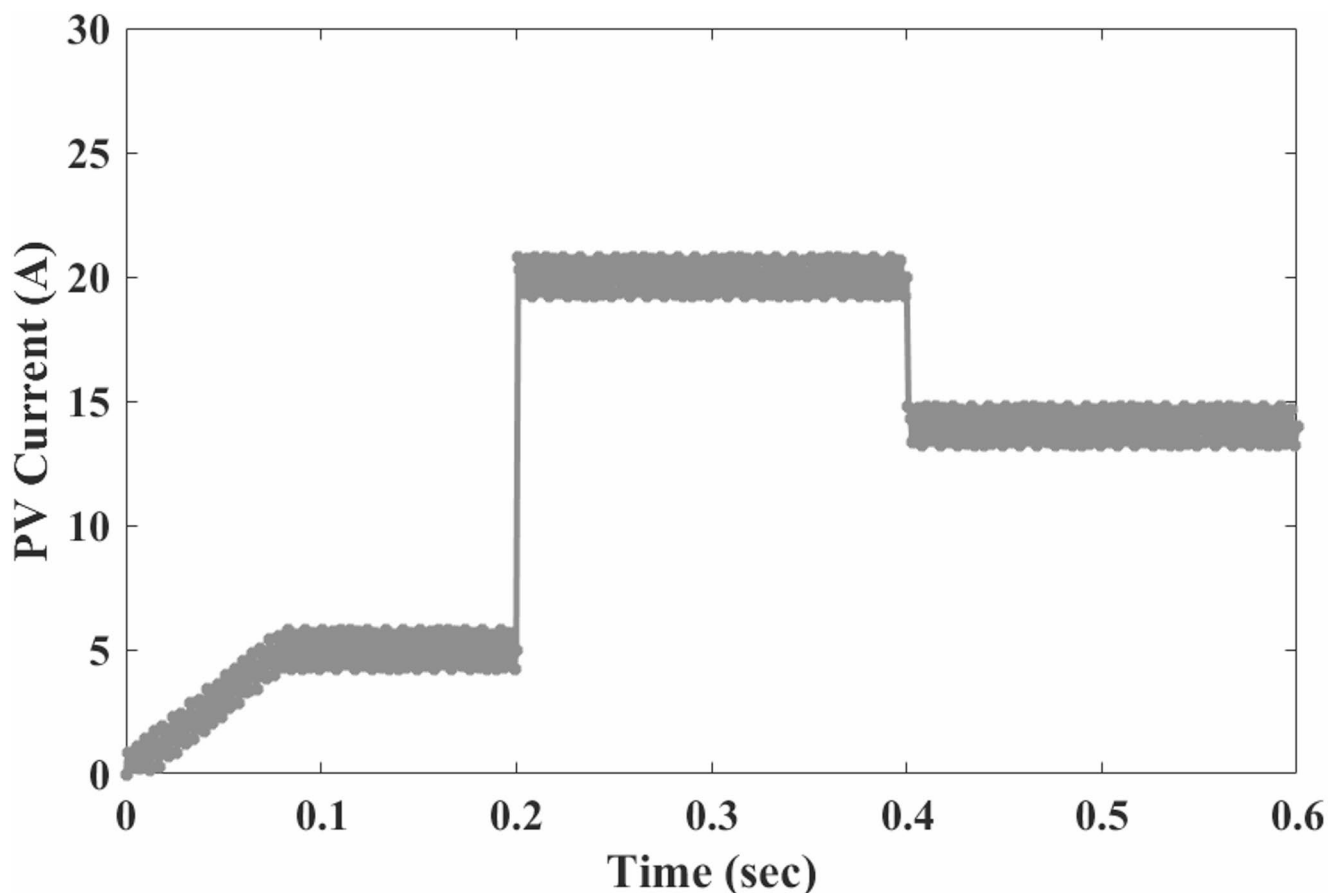


Fig. 9 Analysis of PV current

At a lowest irradiance value of 200 W/m^2 , the torque is found to be 0.18. As a result, only 0.32 speed is reached, whereas the current remains constant at 1.5 A. The torque value is 0.42 at 400 W/m^2 , with a speed of 0.61 and current of 2.8 A. When the solar irradiance rises to 600 W/m^2 , corresponding rise is also observed in torque, speed and current, whose values respectively are 0.61, 0.84 and 3.9 A. For the irradiance of 800 W/m^2 , 0.75 torque and 0.96 speed is reached, with a current of 4.5 A. There is a slight dip in the speed at 1000 W/m^2 , with a torque value of 0.79 and current of 4.7 A. However, at 1200 W/m^2 , torque value is 0.82 with highest speed recorded. Thus, it is essential to monitor the power flow efficiency.

4.7 Efficiency Metrics of the Water Pumping Subsystem under IoT Supervision

The motor performance analysis in terms of water pumping efficiency is presented in Table 3, for different controlling methods like conventional, only MPPT and IoT combined MPPT.

It can be observed from the Table 3, that the combination of IoT and MPPT technique yields better pumping efficiency when the solar irradiance levels vary from 200 W/m^2 to 1000 W/m^2 . At about 400 W/m^2 , the proposed IoT-MPPT model gives an efficiency of 60.4%, while it is only 41.3% with the conventional controlling methods. If 600 W/m^2 is considered, the use of MPPT technique alone raises the efficiency to a good value of 66.9%. This is further lifted to 74.2% on integration of IoT. For 1200 W/m^2 case, owing to presence of abundant solar power, there is a drop in the efficiency. But still, an 83.2% efficiency could be possible with the IoT and MPPT combination as against 65.1% obtained with conventional control schemes. Thus, predictive control is highly active in specific irradiance situations. It is also to be noted that beyond the normal testing environment, there is

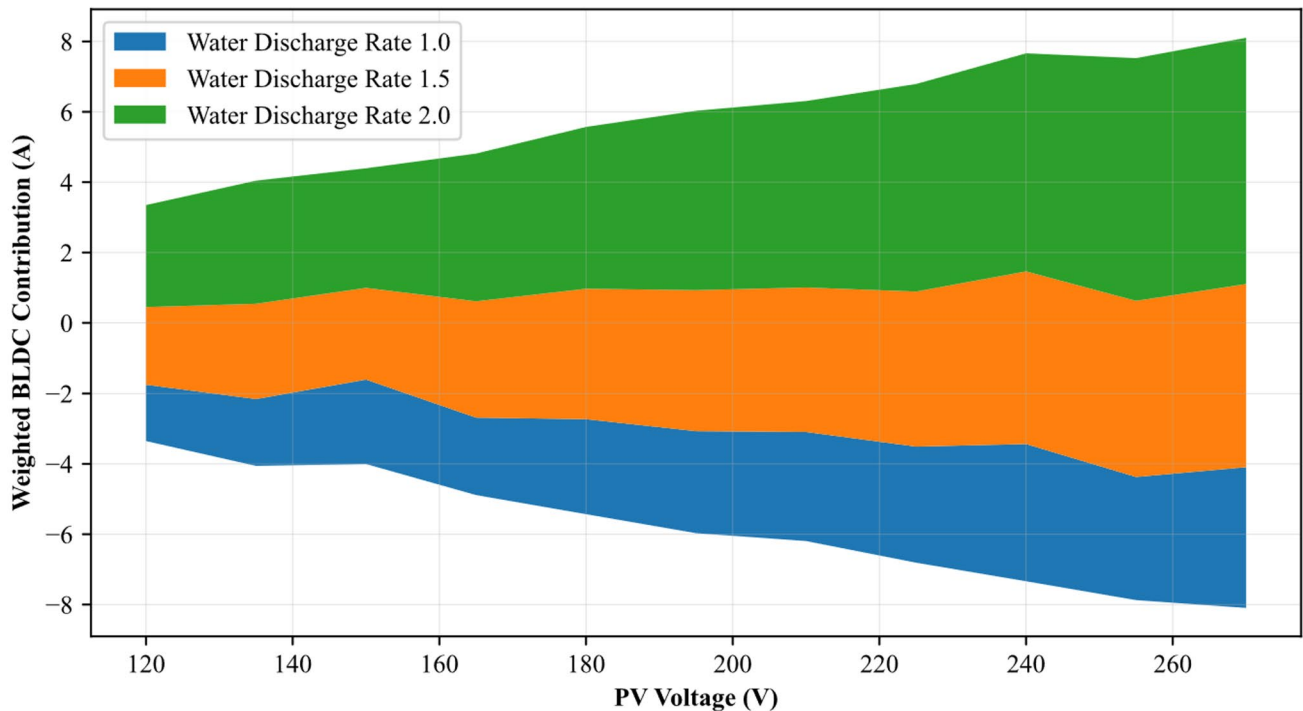


Fig. 10 Real-time system parameter readings: PV voltage, BLDC current and water discharge rate under dynamic conditions

no significant growth in efficiency. There is a peaked efficiency value of 85.5% when incorporating the IoT based MPPT control, during mid-level sunlight, thus underscoring the real-world constraints. To summarize, the use of IoT technique has successfully resulted in efficient conversion of energy to water at all levels. The choice of design factors and operational effects are detailed in the next section.

5 Discussion

Depending on the efficiency pattern observed previously, this section discusses the impacts of various operational parameters in the proposed control system. The performance analysis is carried out for both the time and energy outcomes. Employment of predictive control has exhibited significant gain in the motor operational stability. The effect of discharge dynamics was able to be realized by studying motor and pump interactions. Thus, with the help of these inferences, the proposed system can be compared with common commercial systems on a broader scale.

5.1 Evaluation of Monitoring and Maintenance Strategies

In continuation to the study of motor behaviors under variable conditions, the effect of monitoring strategies in terms of energy utilization and time of response are illustrated in Fig. 14. With traditional control strategies, the utilization of energy is found to be only 58.3% which is below the basic usage expectations. In contrast, the utilization is found to reach 72.6% with the use of IoT based system. Thus, the proposed system almost meets the energy usage targets. As regards the time delay, it is about 11.8 s with the standard control system. With the inclusion of IoT, there is a drastic reduction in time delay to about 5.3 s, which is very significant. This difference underlines the relation of automation with efficiency.

It is also to be noted that the crossing of threshold limits occurs only in the case of IoT control and not with the traditional control. But the response time is very worse in case of conventional control. So, with proper tuning

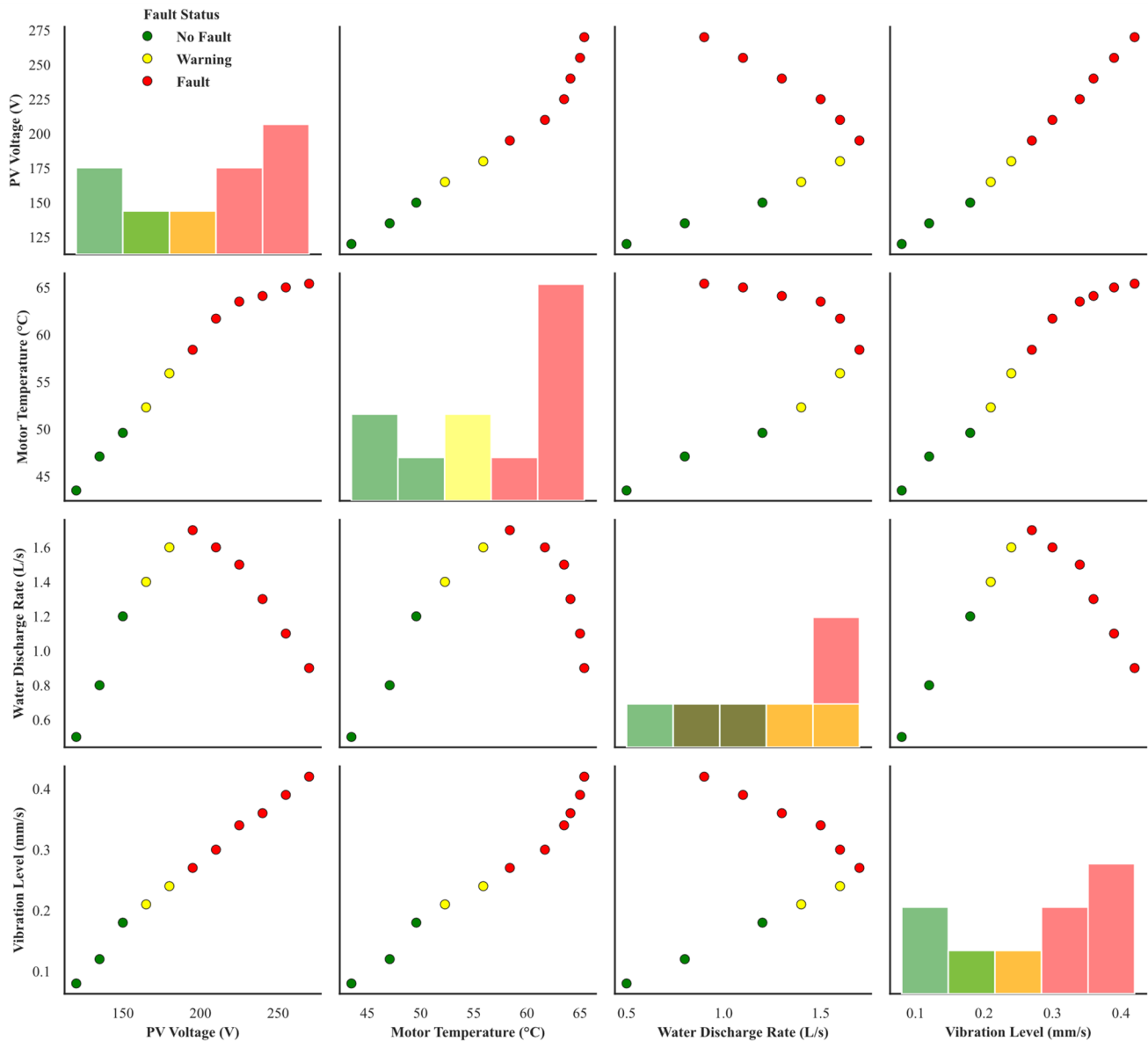


Fig. 11 Fault prediction results from supervised learning models trained on historical sensor data

of threshold limits, IoT system can provide greater stability and control reliability. Thus, with IoT incorporation, there is a potential improvement both in terms of energy usage and responsiveness. The algorithmic support in observing these varying trends is discussed in the subsequent section.

5.2 Impact of Predictive Algorithms on BLDC Motor Operational Stability

Table 4 presents the effect of predictive control mechanism on the stability of BLDC Motor operation.

With the predictive maintenance, the speed ripple drastically dropped to 2.3% from 6.8%, which is obtained without predictive intervention. This has a direct impact on the torque control. There is also a reduction in the torque oscillations from 1.45 Nm to about 0.52 Nm. With the help of proper signal conditioning, it was possible to bring the current fluctuations to 0.9 A from 2.6 A. The Vibration Index with IoT control is very low at about

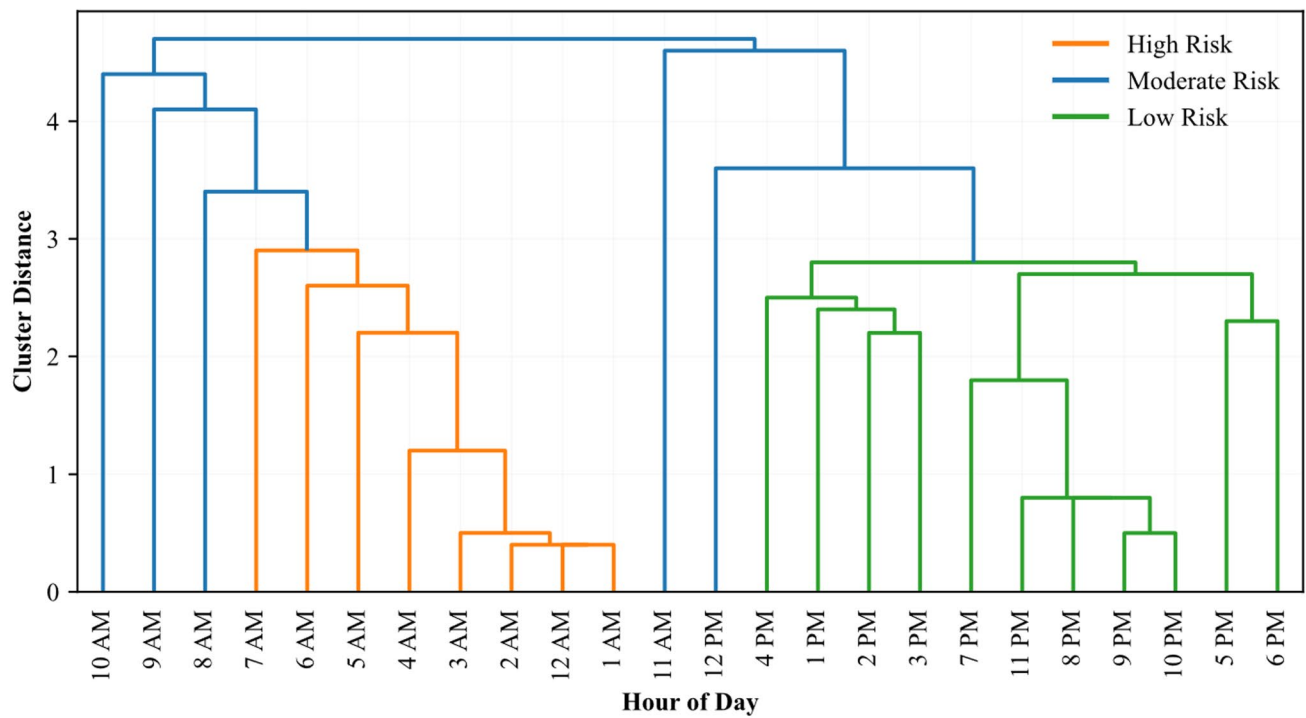


Fig. 12 Optimal maintenance cycle estimation based on failure probabilities and system degradation metrics

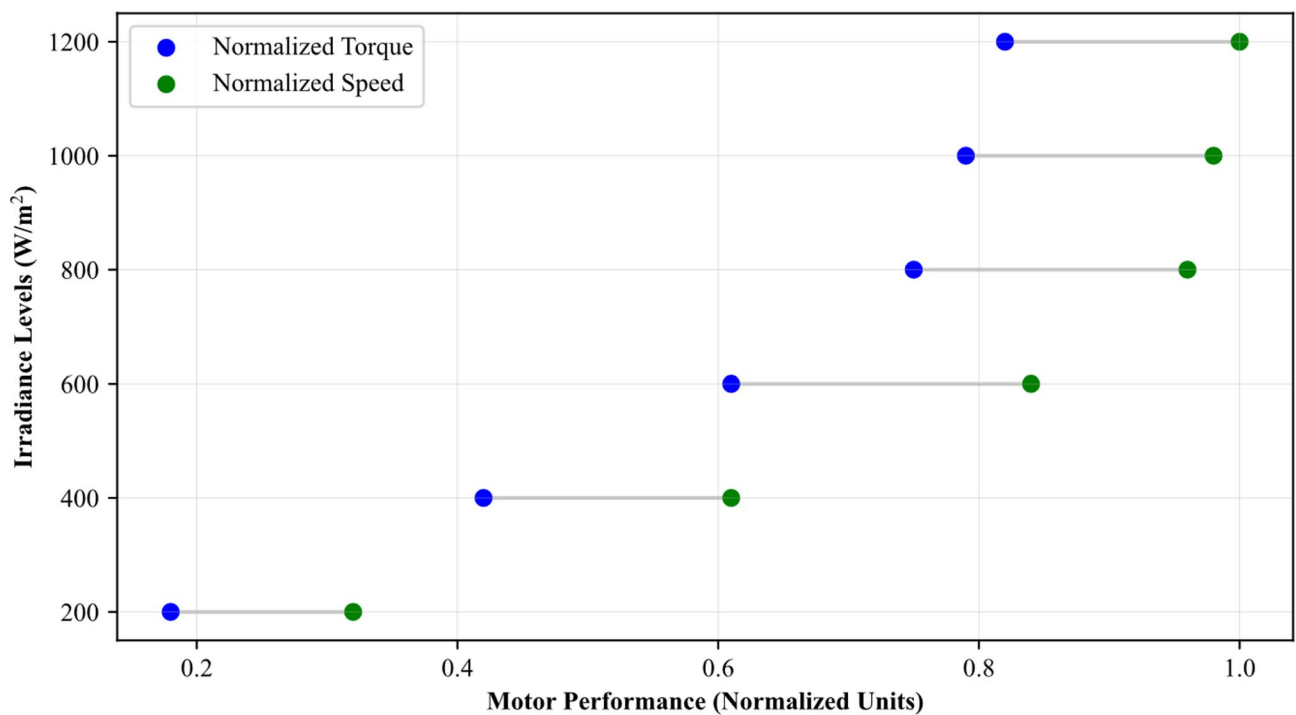
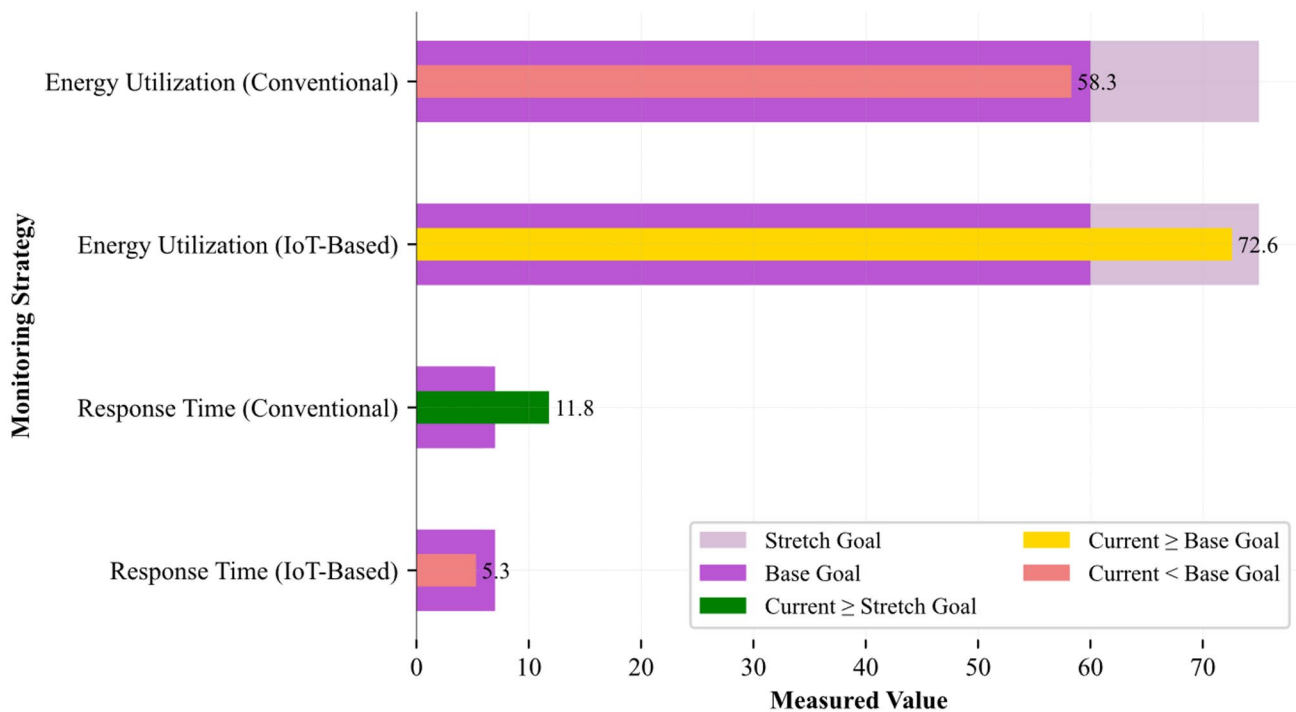


Fig. 13 Motor performance parameters (torque, speed, current) under fluctuating solar irradiance profiles.

Table 3 Water pumping efficiency with various control methods and solar energy levels

Solar Irradiance (W/m ²)	Conventional Control (%)	MPPT Only (%)	IoT+MPPT (%)
200	28.4	34.7	42.1
400	41.3	52.6	60.4
600	56.7	66.9	74.2
800	63.2	72.5	81.9
1000	66.4	78.3	85.5
1200	65.1	76.8	83.2

**Fig. 14** Energy utilization and response time comparison for conventional and IoT-based monitoring strategies**Table 4** Stability indices of BLDC motor operation with and without predictive maintenance intervention

Operating Parameter	Without Predictive Intervention	With Predictive Maintenance
Speed Ripple (%)	6.8	2.3
Torque Oscillation (Nm)	1.45	0.52
Current Fluctuation (A)	2.6	0.9
Vibration Index (mm/s)	3.4	1.1
Transient Recovery Time (ms)	225	87
Overshoot During Load Variation (%)	9.2	3.1
Harmonic Distortion in Back EMF (%)	7.4	2.9
Rotor Deviation Index (rad/s)	0.85	0.27

1.1 mm/s. The transient recovery time was rapid with IoT control, by reducing the time to 87 ms from 225 ms, leading to greater withstanding capability when subjected to disturbances. The waveform quality also improved with a reduction of harmonic distortion from 7.4% to about 2.9%. Lower value of overshoot is observed with the predictive maintenance logic as against the normal system. The reduction in the rotor deviation index to about 0.27 rad/s indicates perfectly aligned mechanical operation. All these improvements are possible with proper tuning of predictive algorithms. The linking of these motor response parameters with that of pump performance are presented in the following section.

5.3 Correlation Between Motor response and Pumping Output

In this section, the focus is on the relation between motor performance and the flow behavior, whose regression analysis is illustrated in Fig. 15.

The hydraulic flow characteristics varies in proportion with the increase in motor load. Denser flow patterns are observed at 125% load, as against 25% load, indicating the correlation between the electrical output and the hydraulic performance. Irrespective of the overlap, the central tendencies stand distinct. With the stabilization of motor control, the flow rate variation narrows down spreading across load bands. From this type of cluster, consistency in the conversion efficiency can be witnessed. In spite of these advantages, there are minor deviations visible from 75% to 100% load bands. With load exceeding 100%, the flow gain becomes a plateau. Till the development of nominal torque, the flow at early stages remains unsatisfactory. In this way, the operation thresholds required for efficient pumping can be determined using regression modeling. The architecture reliability related to these outcomes are detailed in the next section.

5.4 Effectiveness of IoT Architecture on Operational Reliability of Pumping System

Based on the previous discussion about the relation between motor performance and flow rate, the behavior of the IoT supervised system is examined in this section. Figure 16 indicates the role of IoT structure in providing an improved operational uptime. The uptime is found to be increased from about 475 h to 920 h, as against the earlier conventional models. This is made possible through consistent monitoring and adjustive mechanisms. The fault conditions occurring in a traditional system frequently are almost eliminated in the IoT supported model. Further, the use of predictive maintenance algorithm assists in managing the unexpected breakdowns also.

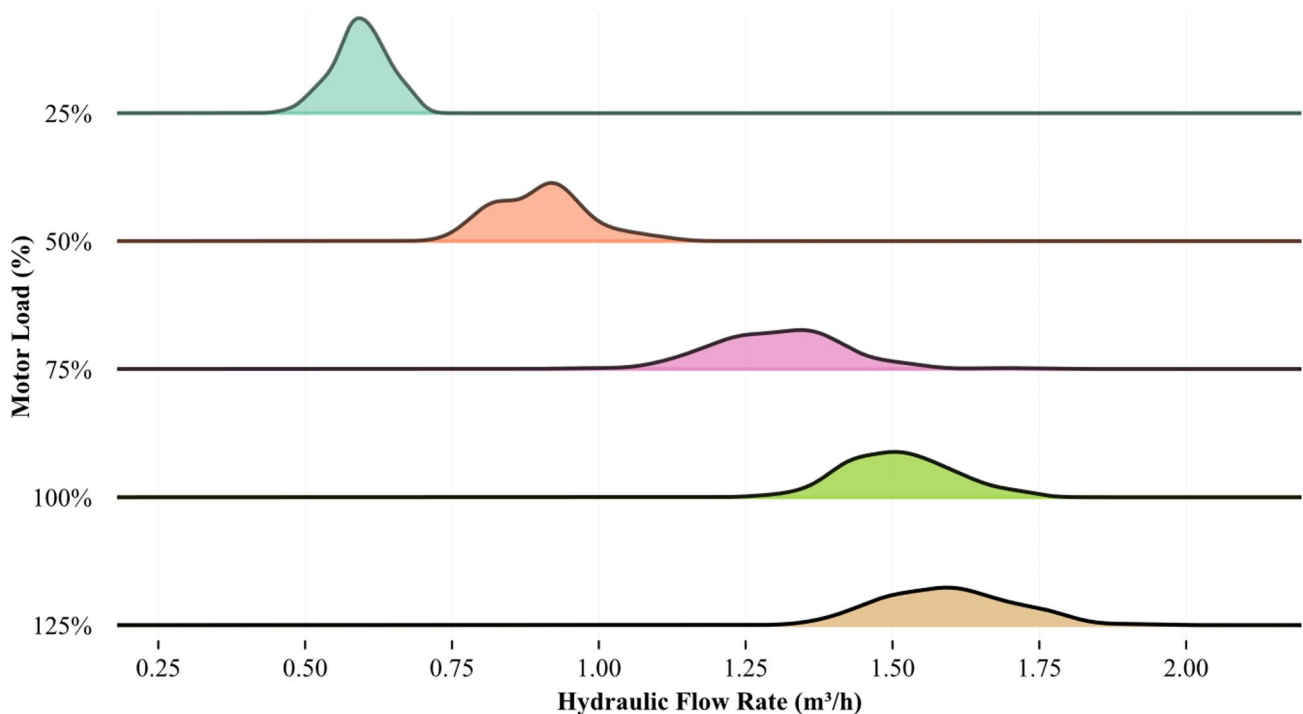


Fig. 15 Regression analysis between motor electrical output and hydraulic flow characteristics

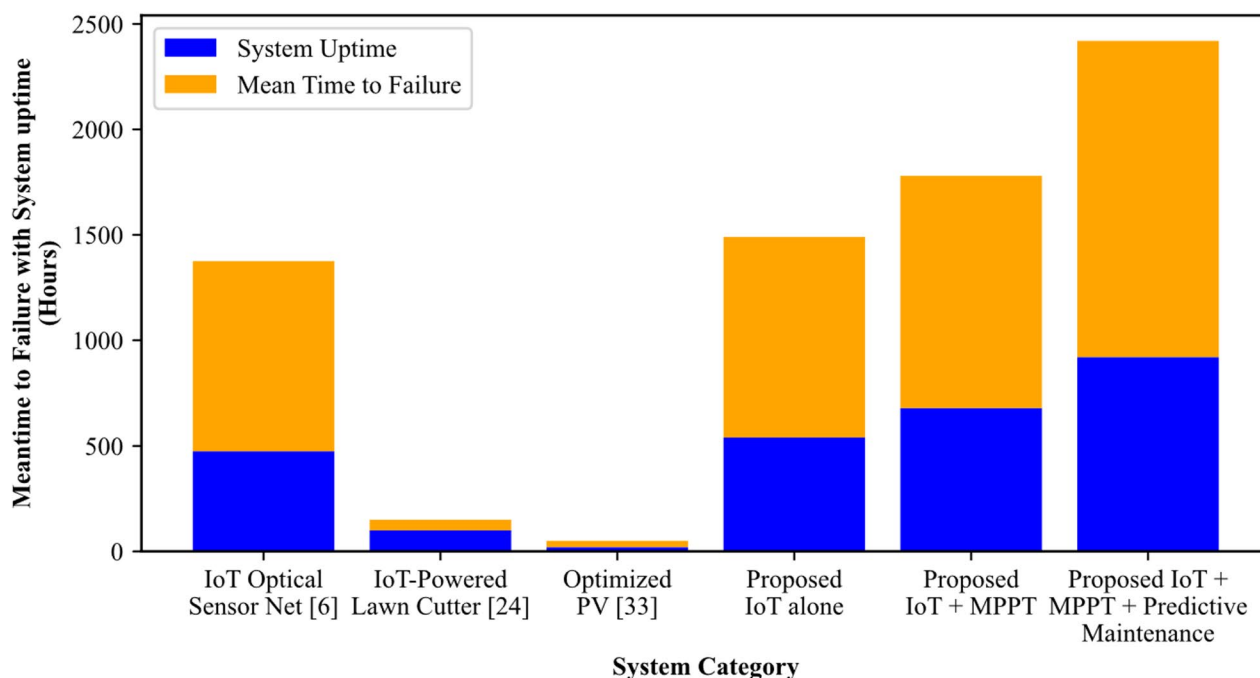


Fig. 16 Mean time to failure and system uptime under IoT supervision

In case of legacy models like [24] exhibiting an uptime of only 100 h, it is possible to achieve 1500 h of uptime before failure, with the proposed framework. Thus, the endurance of the traditional system is greatly amplified. Further, the energy efficiency subjected to varying loads is boosted with application of MPPT. Without the predictive models, the reliable metrics will degrade to a worse value. Hence, system monitoring combined with forecasting is the best way to maintain stability. The architectural design required to meet this objective is presented in this section followed by a comparative study of traditional and proposed system performance. Figure 17 presents the effect of normal and proposed controls in reducing the fault rate.

From Fig. 17, it is clear that the fault detection accuracy is greatly increased with the use of predictive maintenance. This further leads to significant reduction in the occurrence of faults. The proposed IoT supervised Predictive mechanism provides a steeper reliability margin, when compared with the conventional system. Thus, the combination of IoT, MPPT and Predictive arrangement outperforms the other control mechanisms. Even an improved system without any predictive control cannot excel in suppression of faults, thus highlighting the importance of adding predictive control layers in the system architecture. An increase in accuracy is possible only with a decrease in fault rates. To be precise, performance improvement is possible with the change in system design. Thus, the operational resilience can be enhanced with the help of predictive control mechanism.

5.5 Comparative Study with Existing Literature and Commercial Implementations

This section presents comparative performance analysis of the system with and without proposed model implementation. Reference outcomes from various literatures and the commercial research use cases are compared with the proposed system and tabulated in Table 5.

Though systems [1] and [8] exhibit control precision, IoT layers are lacking in their architecture. Systems like [2] focusses only on smart water management but not trading efficiency. Though adaptive logic is used in

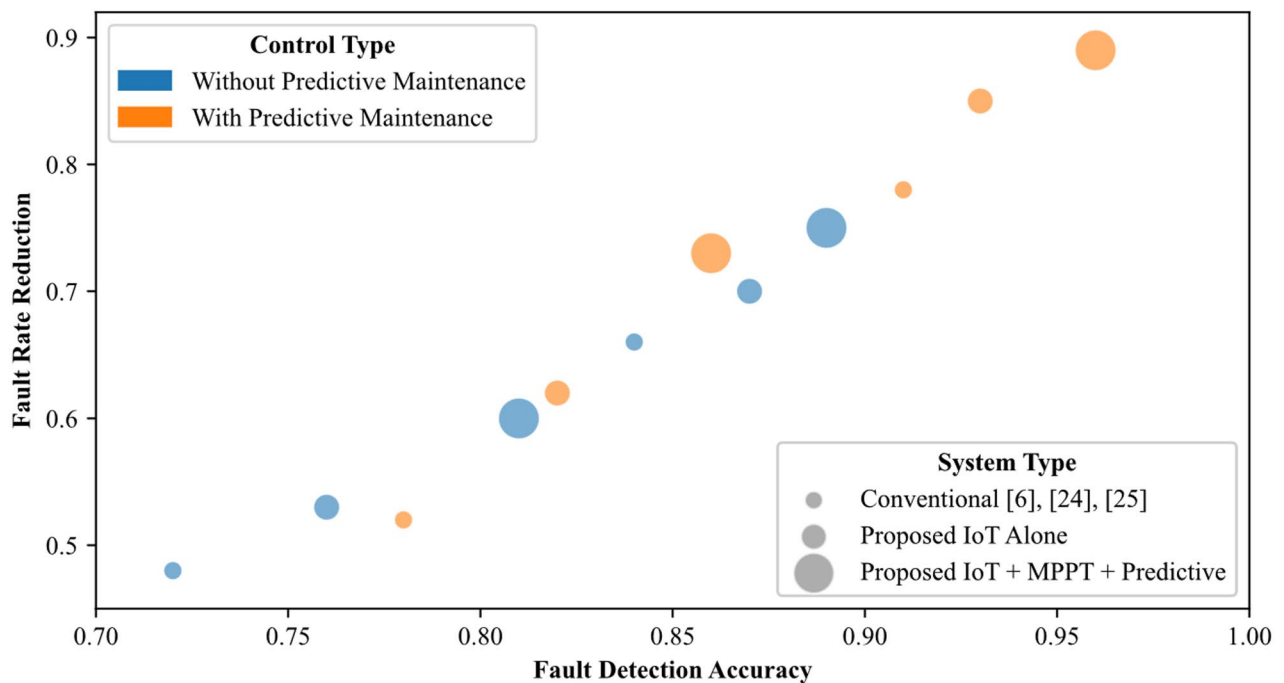


Fig. 17 Comparative fault rate reduction achieved through IoT-based predictive control

Table 5 Benchmarking results of the proposed system against existing PV-fed pumping solutions

Ref. No.	Motor Type	Control Approach	MPPT Accuracy (%)	System Efficiency (%)	Fault Rate Reduction (FRR)	Research Gaps
[1]	BLDC	Optimized MPPT+DC-DC	93.1	84.2	0.58	No IoT integration
[2]	Induction Motor	Smart Water+PV	89	78.5	0.52	Focused on rural domestic supply
[7]	BLDC	Grid-tied BLDC Control	90.2	82.6	0.54	Grid-based operation
[8]	BLDC	NBO-SDRN	92.8	83.7	0.63	Emphasizes control optimization
[10]	DC Shunt	Fuzzy Logic MPPT	87.5	76.4	0.49	Basic optimization, no predictive logic
[13]	BLDC	Dynamic Energy Amendment	91.6	80.1	0.6	Adaptive but lacks IoT layer
[14]	SynRM	Grid-integrated Seamless Ops	90.5	81.3	0.57	Reluctance-based pumping
[16]	DC Motor	High-Performance Control	86	74.9	0.45	Lacks predictive elements
[20]	DC Motor	Chaotic Horse Herd MPPT	94.2	85.1	0.62	Novel optimizer, no IoT
Proposed	BLDC	Hybrid IoT-Predictive Control	96.3	89.4	0.75	Considering all the research gaps

[13], there is no inclusion of predictive modeling [20]. was able to demonstrate 94.2% efficiency using chaotic optimizer, but IoT was not used. As against all these systems, a remarkable 96.3% accuracy is achieved with the proposed system under varying loads. In addition to this, efficiency increased to 89.4% and fault rate reduction to 0.75. These results confirm the enhanced and consistent performance of the system even under dynamic weather conditions and loads. Further, the uncertainties present in the energy dispatch can also be minimized by integrating IoT. To summarize, the various research gaps identified over the literature works are bridged by this proposed setup.

6 Conclusion

The proposed IoT based PV powered BLDC water pumping system coupled with real time monitoring and predictive maintenance features has demonstrated wider benefits to the system operation, by addressing the key issues present in conventional systems. For monitoring the PV powered BLDC pumps in real time, IoT usage and ML-based predictive maintenance models have been suggested. The prediction discrepancies found in the earlier systems are overcome by this method through the use of IoT. Lack of proper monitoring and checking mechanisms are removed in the proposed system, by which steady working is ensured even during weather dynamics. This mixed model enables the system to withstand any type of load changes. Cloud based work helped the system to display the warnings to the user to make them prepared for suitable actions. Thus, there is a greater improvement in the system operation spanning various working metrics. From the results, it is clear that even under varying solar irradiance, the MPPT accuracy was found to be 96.3%. Further, the efficiency of the system has reached 89.4% outperforming other conventional schemes. With fault rate reduction value of 0.75, fault identification and fixing process are greatly enhanced. The response time in fault finding and alarming is also improved to a good extent. In spite of these improvements, there remain few limitations like demand for cloud links to remain on always to get the updates. Since IoT requires digital arrangements, the hardware needs to be modified based on the field location whether rural or urban. In future expansion of the work, this concern can be concentrated by including flexible AI Models and testing the system for different types of farming areas. Integration of this system with smart water systems can also be explored.

Author Contributions S.J – Concepts, Methodology, Manuscript writing R.R – Resources, Simulation K.R – Supervision, Validation R.P.N – Formal analysis, Manuscript reviewing.

Funding No funding is received.

Data Availability The data analyzed are available within the manuscript.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Sevugan Rajesh, J., Karthikeyan, R., Revathi, R.: Analysis and control of grid-interactive PV-fed BLDC water pumping system with optimized MPPT for DC-DC converter. *Sci. Rep.* **14**(1), 25963 (2024)
2. Rumbayan, M., Pundoko, I., Sompie, S.R., Ruindungan, D.G.: Integration of smart water management and photovoltaic pumping system to supply domestic water for rural communities. *Results Eng.* **25**, 103966 (2025)
3. Karthikeyan, S., Gobhinath, S., Sabarish, L., Sabeer, S., Vijay, K., Vignesh, V.V.: Implementation of IoT Based Hybrid Uninterrupted Power Supply. In 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS) (Vol. 1, pp. 705–710). IEEE. (2024), March

4. Thakur, P., Malhotra, M., Bhagat, R.M.: Implementation of an Automated Hydroponic System using ANN: A Case Study on Spinach. In 2023 International Conference on Communication, Security and Artificial Intelligence (ICCSAI) (pp. 341–346). IEEE. (2023), November
5. Sain, C., Banerjee, A., Biswas, P.K., Padmanaban, S.: A state-of-the-art review on solar-powered energy-efficient PMSM drive smart electric vehicle for sustainable development. *Adv. Greener Energy Technol.*, 231–258. (2020)
6. Durai, R.V., Vijayakumar, R., Lakshmisridevi, S., Bhanu, S.T., Arunkumar, U.: IoT Based Optical Sensor Network For Precision Agriculture. In 2024 International Conference on Optimization Computing and Wireless Communication (ICOCWC) (pp. 1–7). IEEE. (2024), January
7. Kar, B.N., Samuel, P., Mallick, A., Pradhan, J.K.: Grid-connected solar PV fed BLDC motor drive for water pumping system, pp. 1477–1504. *Distributed Generation & Alternative Energy Journal* (2023)
8. Joshua, K.P., Rangasamy, L.V., Reddy, C.V.K., Veeruchinnan, R.: Energy management of solar photovoltaic fed water pumping system based BLDC motor drive using NBO–SDRN approach. *Electr. Eng.* **106**(3), 3045–3059 (2024)
9. Singh, V., Gaddala, R.K.: Grid Interactive PV-fed BLDC Motor with Power Quality Features using Predictive Current Control. In 2023 IEEE 2nd Industrial Electronics Society Annual On-Line Conference (ONCON) (pp. 1–6). IEEE. (2023), December
10. Alkuhayli, A., Noman, A.M., Al-Shamma'a, A.A., Abdurraqeeb, A.M., Alharbi, M., Farh, H., H. M., Qamar, A.: Enhancing photovoltaic-powered DC shunt motor performance for water pumping through fuzzy logic optimization. *Machines*. **12**(7), 442 (2024)
11. Sundari, G., Muniraj, R., Shanmugapriyan, J.: Switched reluctance motor based water pumping system powered by solar using hybrid approach. *Appl. Energy*. **365**, 123188 (2024)
12. Errouha, M., Combe, Q., Ouanjli, N.E., Motahhir, S.: A review of modern techniques for efficient control of AC motors utilized in PV water pumping system. *Irrig. Sci.*, 1–21. (2024)
13. Arun Kumar, U., Chandraguptamauryan, K.S., Chandru, K., Babu, R.: W. A novel dynamic energy amendment control for solar fed brushless DC motor for water pumping system. In *Inventive Systems and Control: Proceedings of ICISC 2023* (pp. 253–266). Singapore: Springer Nature Singapore. (2023)
14. Parveen, H., Singh, B.: PV Array fed synchronous reluctance motor driven water pumping system with grid integration and seamless operability. *IEEE Trans. Ind. Appl.* **60**(2), 3724–3733 (2024)
15. Kumar, A., Marwaha, S., Manna, M.S., Marwaha, A., Kumar, R., Amir, M., Zaitsev, I.: Comparative analysis of brushless DC and switched reluctance motors for optimizing off-grid water pumping. *Sci. Rep.* **15**(1), 3527 (2025)
16. Talbi, B., Krim, F., Rekioua, T., Mekhilef, S., Laib, A., Belaout, A.: A high-performance control scheme for photovoltaic pumping system under sudden irradiance and load changes. *Sol. Energy*. **159**, 353–368 (2018)
17. Angadi, S., Yaragatti, U.R., Suresh, Y., Raju, A.B.: Comprehensive review on solar, wind and hybrid wind-PV water pumping systems-an electrical engineering perspective. *CPSS Trans. Power Electron. Appl.* **6**(1), 1–19 (2021)
18. Parveen, H., Sharma, U., Singh, B.: Pole reduction concept for control of synchronous reluctance motor-based solar photovoltaic water pumping system for improved performance. *IEEE Trans. Industr. Electron.* **68**(7), 5650–5660 (2020)
19. Bjaoui, M., Larroque, B., Djebali, R., Khemissi, L., Ruscassié, R., Benadli, R., Sellami, A.: Three-Phase PV Pumping System With Advanced Control for Enhanced Efficiency and Robustness: Modeling, Experimental Validation, and Optimization. *International Journal of Circuit Theory and Applications* (2025)
20. Abbassi, R., Saidi, S.: Design of a novel chaotic horse herd optimizer and application to MPPT for optimal performance of stand-alone solar PV water pumping systems. *Mathematics*. **12**(4), 594 (2024)
21. Jain, R.K., Barry, V.R., Gadiraju, H.K.V.: Model-based design and sliding mode control approach for two-stage water pumping system with reduced sensors. *IEEE J. Emerg. Sel. Top. Power Electron.* **10**(4), 3940–3949 (2021)
22. Sharma, U., Singh, B.: Utility-tied solar water pumping system for domestic and agricultural applications. *J. Institution Eng. (India): Ser. B*. **101**, 79–91 (2020)
23. Narendra, A., Panda, A.K., Lenka, R.K.: Solar PV fed FSVSI based Variable Speed IM Drive using ASVM Technique. *Eng. Sci. Technol. Int. J.* **40**, 101366 (2023)
24. Suganya, T., Mangaiyarkarasi, P., Thirugnanam, G., Sathish Kumar, T.M.: A hybrid approach with MPPT controller for weed cutting based on solar powered lawnmower with minimal intervention of human involvement adopting IoT technology. *Analog Integr. Circuits Signal Process.* **119**(2), 249–267 (2024)
25. Gan, X., Pei, J., Pavesi, G., Yuan, S., Wang, W.: Application of intelligent methods in energy efficiency enhancement of pump system: A review. *Energy Rep.* **8**, 11592–11606 (2022)
26. Hazil, O., Allouani, F., Bououden, S., Chadli, M., Chemachema, M., Boulkaibet, I., Neji, B.: A robust model predictive control for a photovoltaic pumping system subject to actuator saturation Nonlinearity. *Sustainability*. **15**(5), 4493 (2023)
27. Kavya, V., Jegathesan, V., Jarin, T.: A novel approach for solar combined open-ended winding induction machine for agriculture water pumping applications, vol. 33, p. 101141. *Sensors, Measurement* (2024)
28. Mani, P., Mahadevan, S.K., Johnson, A.R., Kullani, M.: An optimized design modelling of PV integrated SEPIC-based four-switch inverter for sensorless PMBLDC motor control. *Automatika*. **63**(1), 90–101 (2022)
29. Jones Chullai, E.R., Nannam, H., Roy, P., Roy, R., Banerjee, A.: Performance evaluation of a TOFOSMC-based pump hydro energy storage in the application of grid-connected photovoltaic system. *Electr. Eng.* **107**, (8), 11121–11142 (2025)

30. Shukla, S., Singh, B.: MRAS based speed estimation of single stage solar powered vector controlled induction motor drive for water pumping. In 2016 IEEE 7th Power India International Conference (PIICON) (pp. 1–6). IEEE. (2016), November
31. Elymany, M.M., Enany, M.A., Metwally, H., Shaier, A.A.: Enhanced operation of PVWPS based on advanced soft computing optimization techniques. *Sci. Rep.* **14**(1), 29429 (2024)
32. Verma, A.D., Gaur, P., Tomar, A.: Secondary Droop Control Supported Cascaded PR Controller for SCIM-Based Water Pumping System. In 2024 IEEE 4th International Conference on Sustainable Energy and Future Electric Transportation (SEFET) (pp. 1–6). IEEE. (2024), July
33. Dutta, N., Kaliannan, P., Shanmugam, P.: Application of machine learning for inter turn fault detection in pumping system. *Sci. Rep.* **12**(1), 12906 (2022)
34. Selvan, S.S.: Maximum power point tracking for grid tied solar fed DTC controlled IM drive using artificial neural network with energy management. *Int. J. Environ. Technol. Manage.* **27**(1–2), 151–172 (2024)
35. Mahmoud, Z.B., Khedher, A.: On the common mode current reduction in induction motor fed by three-level NPC inverter for pumping photovoltaic application. *Int. J. Power Electron.* **21**(1), 28–45 (2025)
36. Mishra, A.K., Singh, B.: An efficient control scheme of grid supported 4-phase switched reluctance motor-driven SPWPS. *IEEE Trans. Energy Convers.* **35**(3), 1258–1267 (2020)
37. Senpinar, A.: Internet-/Arduino-controlled PV automatic irrigation system for clean environment. *Int. J. Environ. Sci. Technol.* **16**, 5185–5196 (2019)
38. Sayed, B.T., Rodrigues, P., Chammam, A., Kareem, A.K., Trivedi, T., Singh, A.: A Bayesian machine learning-driven control algorithm for one-phase winding permanent-magnet brushless DC machines. *Int. J. Dynamics Control.* **13**(1), 1–13 (2025)
39. Belghiti, H., Kandoussi, K., Harrison, A., Moustaine, F.Z., Otmani, R.E., Sadek, E.M., Dost Mohammadi, S.A.: A novel adaptive FOCV algorithm with robust IMRAC control for sustainable and high-efficiency MPPT in standalone PV systems: experimental validation and performance assessment. *Sci. Rep.* **14**(1), 31962 (2024)
40. Thangavel, J., Chinnaraj, G., Chandrasekaran, G., Kumarasamy, V.: Design and development of solar photovoltaic fed modular multilevel inverter using intelligent techniques for renewable energy applications. *J. Intell. Fuzzy Syst.* **44**(2), 1807–1821 (2023)
41. Porselvi, T., Krithika, K., Ganesh, C.S.: PV system based induction motor with landsman converter using IFOC controller. In 2021 2nd global conference for advancement in technology (GCAT) (pp. 1–4). IEEE. (2021), October
42. Arsad, A.Z., Azhar, A.D., Arsad, S.R., Zuhdi, A.W.M., Chau, C.F., Ghazali, A.: Towards artificial intelligence for solar charge controller: an analytical study of recent status, optimization and module development. *Energ. Syst.*, 1–54. (2025)
43. Kumar, A., Dubey, A.K., Segovia Ramirez, I., del Muñoz, A., García Márquez, F.P.: Artificial intelligence techniques for the photovoltaic system: A systematic review and analysis for evaluation and benchmarking, pp. 1–25. *Archives of Computational Methods in Engineering* (2024)
44. Reegan, J., Gnana Saravanan, A.: Optimized PI Control for PV-Powered PMBLDC Motor with SEPIC-Zeta Converter. *J. Electr. Eng. Technol.* **19**(7), 4215–4237 (2024)
45. Ismael, D.M., Atyia, T.H.: Comparative evaluation of speed control techniques for switched reluctance motors in industrial applications. *Eng. Res. Express.* **6**(3), 032304 (2024)
46. Naresh, K., Suresh, D., Pattnaik, S.: High-Voltage Gain DC–DC Converter for Sensorless Speed Control of BLDC Motor. *International Journal of Circuit Theory and Applications* (2025)
47. Pal, S., Bhattacharya, A.: Open-Loop Capacitor Voltage Balancing of a Nine-Level Packed U-Cell Converter. *IEEE Trans. Ind. Appl.* **60**(4), 6446–6457 (2024)
48. Guerrero-Ramirez, E., Martinez-Barbosa, A., Contreras-Ordaz, M.A., Guerrero-Ramirez, G., Guzman-Ramirez, E., Barahona-Avalos, J.L., Adam-Medina, M.: DC motor drive powered by solar photovoltaic energy: An FPGA-based active disturbance rejection control approach. *Energies.* **15**(18), 6595 (2022)
49. Dhaked, D.K., Narayanan, V.L., Gopal, R., Sharma, O., Bhattarai, S., Dwivedy, S.K.: Exploring deep learning methods for solar photovoltaic power output forecasting: A review. *Renew. Energy Focus.* **53**, 100682 (2025)
50. Narayanan, V.L., Dhaked, D.K., Sitharthan, R.: Improved machine learning-based pitch controller for rated power generation in large-scale wind turbine. *Renew. Energy Focus.* **50**, 100603 (2024)
51. Dhaked, D.K., Kumar, P., Ganguly, S.: Development of data driven model for proton exchange membrane fuel cell using machine learning approaches. In 2024 IEEE 3rd International Conference on Control, Instrumentation, Energy & Communication (CIEC) (pp. 67–72). IEEE. (2024), January
52. Sharma, P., Dhaked, D.K., Sharma, A.K.: Mathematical modeling and simulation of DC-DC converters using state-space approach. In *Proceedings of Second International Conference on Smart Energy and Communication: ICSEC 2020* (pp. 11–29). Singapore: Springer Singapore. (2021), January
53. Singh, M., Yadav, M.R., Dhaked, D.K.: A review on various voltage boosting topology in DC-DC converter. *Recent. Adv. Electr. Electron. Eng. (Formerly Recent. Pat. Electr. Electron. Engineering).* **17**(6), 554–572 (2024)
54. Sailaja, K.I., Rahimunnisa, K.: Analysis of energy management in a hybrid renewable power system using MOA technique. *Environ. Dev. Sustain.* **26**(7), 18989–19011 (2024)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Sevugan Rajesh. J¹ · Revathi. R² · Karthikeyan. R³ · Rajnarayanan. P. N⁴

✉ Sevugan Rajesh. J
sevugan.rajesh123@gmail.com

Revathi. R
revthir@gmail.com

Karthikeyan. R
papkarthik@gmail.com

Rajnarayanan. P. N
pnrajnarayanan@gmail.com

¹ Department of ECE, Dr.NGP Institute of Technology, Coimbatore, India

² Department of EEE, KPR Institute of Engineering and Technology, Coimbatore, India

³ Department of EEE, M.Kumarasamy College of Engineering, Karur, India

⁴ Department of EEE, Theni Kammavar Sangam College of Technology, Theni, India